True or false?

A temperature that is 7°C at t = 0, 23°C at t = 12 h, and repeats every 24 hours is modeled by $15 - 8\cos(\pi t/12)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q3/5

A blood pressure that starts at 100 mm Hg and rises, varies between 80 and 120, and repeats every 0.8 s is modeled by

A. $100 + 20\sin(0.8t)$

B. $20 + 100\sin(2.5\pi t)$

C. $120\sin(2.5\pi t)$

D. $100 + 20\sin(2.5\pi t)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q4/5

A piston's position starts at 0 cm and increases, varies between −3 and 3 cm, and repeats every 1 s. Write a model for p.

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q5/5

SET 44 • HSF-IF.B.5 • INTERPRETING FUNCTIONS

State domains of radical, rational, logarithmic, and polynomial models

MAIN	Algebra II main question	**HSF-IF.B.5**
BACKWARD 1	Domains of linear, quadratic, and exponential models	**HSF-IF.B.5**
BACKWARD 2	Division by zero	**7.NS.A.2.b**
BACKWARD 3	Solve linear inequalities	**HSA-REI.B.3**
FORWARD 1	Domains defined by nonlinear conditions	**HSF-IF.B.5**
FORWARD 2	Domains and ranges of inverse functions	**HSF-BF.B.4.a**

SET 44 • HSF-IF.B.5

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5

A car's speed is $s(d) = \sqrt{(22.5d)}$ mph for a skid of d feet. Give the domain of the function and the domain that makes sense in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q1/5

The average cost of x shirts is A(x) = (500 + 4x)/x. Give the domain of the function and the domain in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q2/5

$pH = -\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration. What is the domain?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q3/5

A box from a 12-in sheet has volume $V(x) = x(12 - 2x)^2$. Give the domain of the polynomial and the domain in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q4/5

Give the domain of $f(x) = \sqrt{(x - 2)/(x - 5)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 1

Domains of linear, quadratic, and exponential models • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

$h(t) = -16t^2 + 64t$ models height until the ball lands. Give the domain in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q1/5

C(n) = 2n + 5 is the cost of n tickets. Give the domain in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q2/5

True or false?

The domain of $f(x) = 2^x$ is $x > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q3/5

A model for the number of people uses a domain of

A. fractions only

B. all real numbers

C. whole numbers

D. negative numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q4/5

Why might a model's domain be smaller than the function's domain?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 2

Division by zero • Grade 7 • The Number System • 7.NS.A.2.b

For which x is $1/(x - 5)$ undefined?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q1/5

For which x is (500 + 4x)/x undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q2/5

True or false?

$3/0 = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q3/5

$(x + 1)/(x^2 - 9)$ is undefined at

A. $x = 0$

B. $x = 9$

C. $x = -1$

D. $x = \pm 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q4/5

Is 0/5 defined? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 3

Solve linear inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $x - 2 \geq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q1/5

Solve $12 - 2x > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q2/5

True or false?

$22.5d \geq 0$ means $d \geq 0$.

A. True

B. False

$3 - x \geq 0$ means

A. $x \geq 3$

B. $x \leq 3$

C. $x \geq -3$

D. $x \leq -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q4/5

Solve $2x + 8 > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • HSF-IF.B.5

FORWARD 1

Domains defined by nonlinear conditions • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $f(x) = \sqrt{(9 - x^2)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q1/5

Give the domain of $g(x) = \log(x^2 - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q2/5

Give the domain of $h(x) = 1/(x^2 - x - 6)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q3/5

True or false?

The domain of $\sqrt{(x^2)}$ is $x \geq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q4/5

The domain of $\log(x - 1) + \sqrt{(5 - x)}$ is

A. $1 < x \leq 5$

B. $x \leq 5$

C. $1 \leq x \leq 5$

D. $x > 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSF-IF.B.5

FORWARD 2

Domains and ranges of inverse functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = \sqrt{(x - 2)}$ has domain $x \geq 2$ and range $y \geq 0$. Write f^{-1} and give its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q1/5

$f(x) = 2^x$ has range $y > 0$. What is the domain of $f^{-1}(x) = \log_2 x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q2/5

True or false?

The domain of an inverse function is the range of the original function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q3/5

$f(x) = 1/x + 3$ $(x \neq 0)$ has an inverse with domain

A. $x \neq 0$

B. $x > 3$

C. all real x

D. $x \neq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q4/5

Explain why the inverse of $s(d) = \sqrt{(22.5d)}$ needs the restriction $s \geq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q5/5

SET 45 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Calculate and interpret average rates of change of nonlinear models

MAIN	Algebra II main question	**HSF-IF.B.6**
BACKWARD 1	Average rate of change of simpler functions	**HSF-IF.B.6**
BACKWARD 2	Slope between two points	**8.F.B.4**
BACKWARD 3	Evaluate nonlinear functions	**HSF-IF.A.2**
FORWARD 1	Compare average rates on different intervals	**HSF-IF.B.6**
FORWARD 2	Compare rates of change across representations	**HSF-IF.C.9**

SET 45 • HSF-IF.B.6

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6

A culture has $N(t) = 200 \cdot 2^t$ bacteria after t hours. Find the average rate of change from t = 1 to t = 3 and interpret it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q1/5

A stone's height is $h(t) = 64 - 16t^2$ ft. Find the average rate of change from $t = 0.5$ to $t = 1.5$ s and interpret it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q2/5

The table shows a car's value. Find the average rate of change from t = 2 to t = 4 years and interpret it.

t (years)	0	1	2	3	4	5
V ($)	20000	17000	14450	12282.50	10440.06	8874.11

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q3/5

Speed from skid length is $s(d) = \sqrt{(24d)}$ mph. Find the average rate of change from $d = 24$ to $d = 96$ ft and interpret it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q4/5

Temperature is $T(t) = 15 - 8\cos(\pi t/12)$ °C after t hours. Find the average rate of change from $t = 0$ to $t = 6$ and interpret it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 1

Average rate of change of simpler functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$. Find the average rate of change from $x = 1$ to $x = 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q1/5

A table gives f(2) = 10 and f(5) = 1. Find the average rate of change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q2/5

True or false?

The average rate of change from a to b is $(f(b) - f(a))/(b - a)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q3/5

The average rate of change of f(x) = 3x + 1 on any interval is

A. it varies

B. 1

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q4/5

What are the units of a rate if f is in dollars and x is in years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 2

Slope between two points • Grade 8 • Functions • 8.F.B.4

Find the slope between (1, 400) and (3, 1600).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q1/5

Find the slope between (0.5, 60) and (1.5, 28).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q2/5

True or false?

A negative slope means the output increases as the input increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q3/5

The slope between (24, 24) and (96, 48) is

A. 72/24

B. 1/3

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q4/5

Find the slope between (0, 7) and (6, 15).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 3

Evaluate nonlinear functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find $N(3)$ for $N(t) = 200 \cdot 2^t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q1/5

Find $h(1.5)$ for $h(t) = 64 - 16t^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(24 \cdot 96)} = 2304$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q3/5

$15 - 8\cos(\pi/2)$ equals

A. 15

B. 7

C. 23

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q4/5

Find $T(0)$ for $T(t) = 15 - 8\cos(\pi t/12)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q5/5

SET 45 • HSF-IF.B.6

FORWARD 1

Compare average rates on different intervals • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6

For $N(t) = 200 \cdot 2^t$, compare the average rates on [0, 1] and [3, 4]. What does the comparison show?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q1/5

For $s(d) = \sqrt{(24d)}$, compare the average rates on $[0, 24]$ and $[96, 150]$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q2/5

True or false?

For a function with period 24, the average rate of change over [0, 24] is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q3/5

For $f(x) = x^3$, the average rate on $[-1, 1]$ is

A. 3

B. 0

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q4/5

Find an interval of length 6 on which $T(t) = 15 - 8\cos(\pi t/12)$ has average rate of change $-4/3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSF-IF.B.6

FORWARD 2

Compare rates of change across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

$f(t) = 200 \cdot 2^t$, and g is given by $g(1) = 500$, $g(3) = 1500$. Which has the greater average rate on $[1, 3]$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q1/5

A graph of a cooling liquid falls from 90°C at t = 0 to 50°C at t = 20 min. Compare with T(t) = 90 − 1.5t on the same interval.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q2/5

True or false?

A function with a greater average rate on one interval must be greater everywhere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q3/5

A table shows $h(0) = 0$, $h(2) = 40$; the formula $k(t) = 5t^2$ gives

A. no rate for k

B. a greater rate for k

C. a greater rate for h

D. equal average rates on [0, 2]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q4/5

Explain how to compare rates when one function is a graph and one is a formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q5/5

SET 46 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph square root and cube root functions with transformations

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Graph $y = \sqrt{x}$ and $y = \sqrt[3]{x}$ from a table	**HSF-IF.C.7.b**
BACKWARD 2	Translations and reflections of graphs	**HSF-BF.B.3**
BACKWARD 3	Square and cube roots of numbers	**8.EE.A.2**
FORWARD 1	Graph radical functions with horizontal stretches and reflections	**HSF-IF.C.7.b**
FORWARD 2	Graph a radical function and its inverse	**HSF-BF.B.4.a**

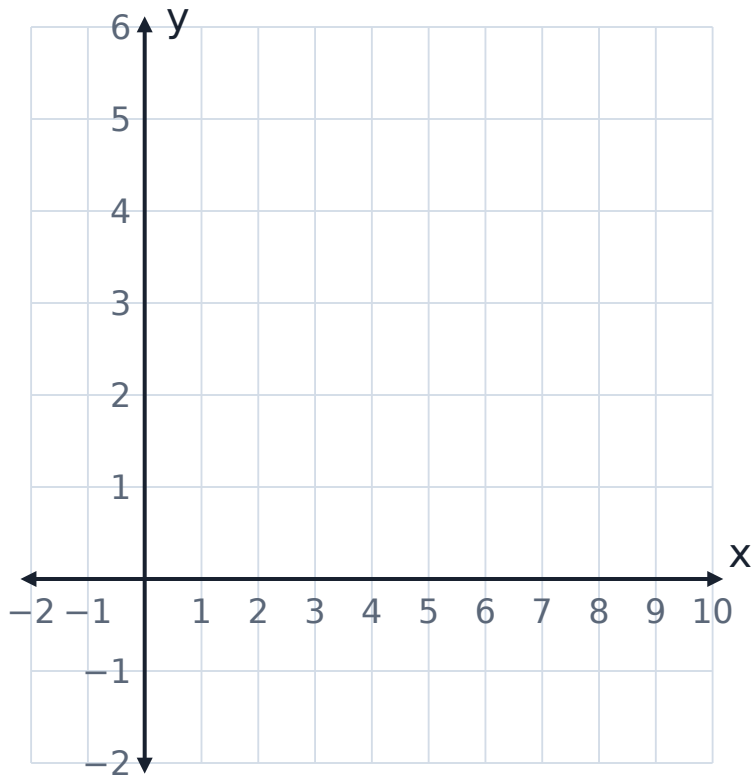
SET 46 • HSF-IF.C.7.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph the function on the grid. Show and label the starting point and two more points, and state the domain and range.

$$y = \sqrt{(x - 2)} + 1$$



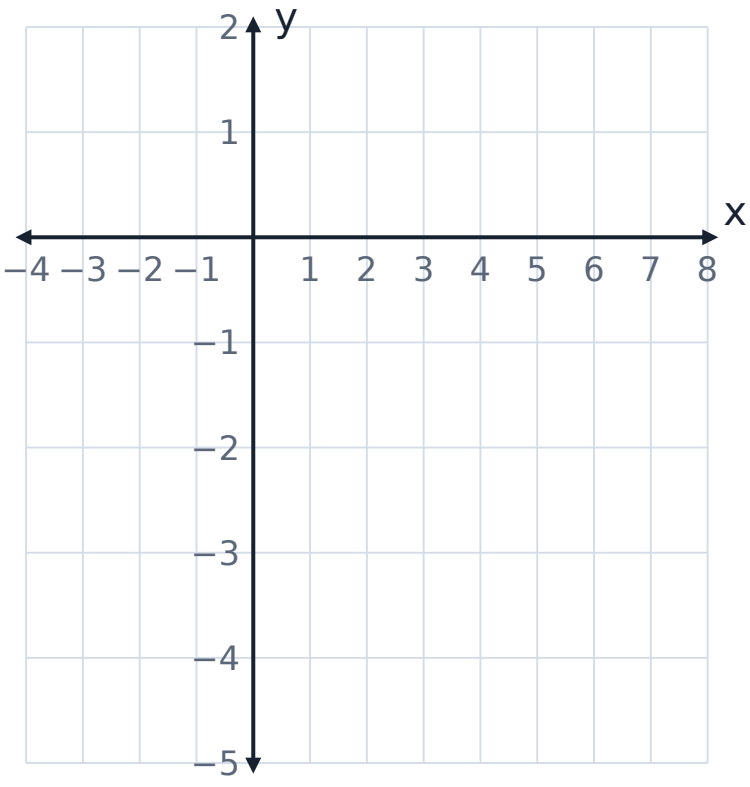
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q1/5

Graph the function on the grid. Show and label the starting point and two more points, and state the domain and range.

$y = -\sqrt{(x + 3)}$



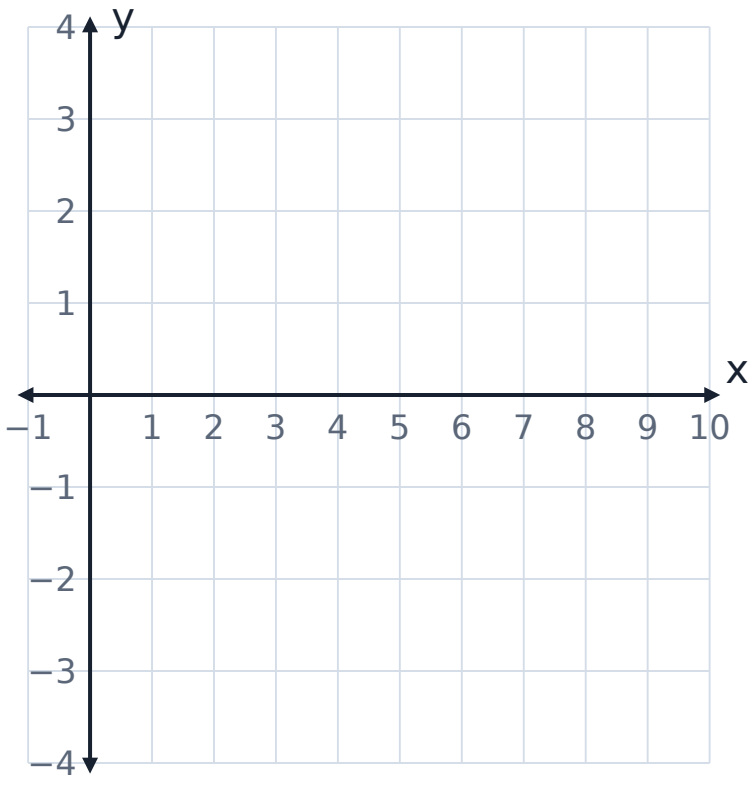
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q2/5

Graph the function on the grid. Show and label the starting point and two more points, and state the domain and range.

$y = 2\sqrt{x} - 3$



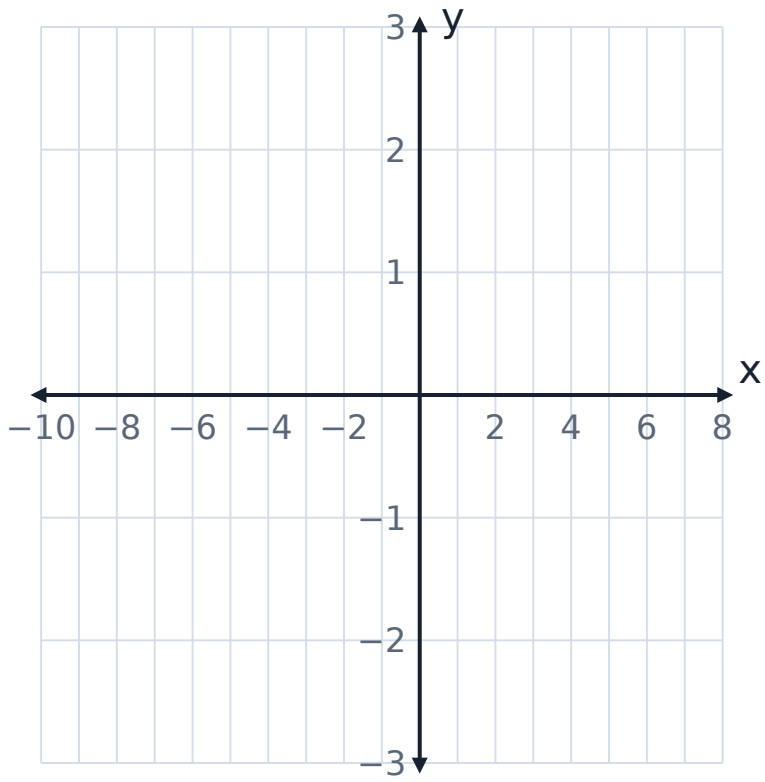
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q3/5

Graph the function on the grid. Show and label the center point and two points on each side, and state the domain and range.

$y = \sqrt[3]{(x + 1)}$



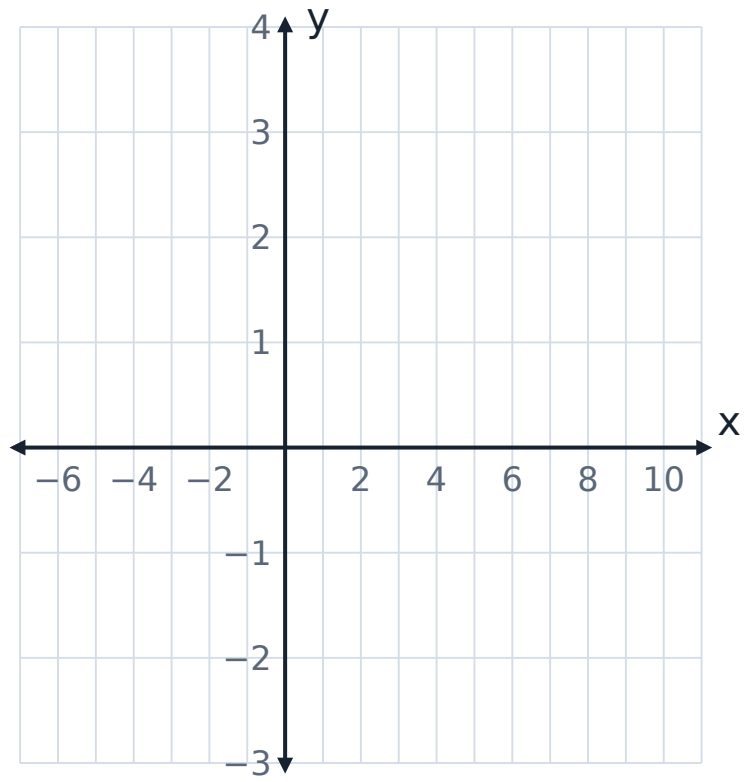
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q4/5

Graph the function on the grid. Show and label the center point and two more points, and state the domain and range.

$$y = -\sqrt[3]{(x - 2)} + 1$$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 1

Graph $y = \sqrt{x}$ and $y = \sqrt[3]{x}$ from a table • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

Complete the table for $y = \sqrt{x}$ at $x = 0, 1, 4, 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q1/5

Complete the table for $y = \sqrt[3]{x}$ at $x = -8, -1, 0, 1, 8$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q2/5

True or false?

The graph of $y = \sqrt{x}$ starts at the origin and decreases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q3/5

The domain of $y = \sqrt[3]{x}$ is

A. $x > 0$

B. all real numbers

C. $x \leq 0$

D. $x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q4/5

Give the range of $y = \sqrt{x}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 2

Translations and reflections of graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is $y = \sqrt{(x - 2)} + 1$ obtained from $y = \sqrt{x}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q1/5

How is $y = -\sqrt{(x + 3)}$ obtained from $y = \sqrt{x}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q2/5

True or false?

$y = 2\sqrt{x}$ is a vertical stretch of $y = \sqrt{x}$ by 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q3/5

The point (0, 0) of $y = \sqrt{x}$ moves to which point for $y = \sqrt{(x - 2)} + 1$?

A. (2, 1)

B. (1, 2)

C. (−2, 1)

D. (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q4/5

Where does (1, 1) of $y = \sqrt[3]{x}$ go under $y = \sqrt[3]{(x + 1)}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 3

Square and cube roots of numbers • Grade 8 • Expressions & Equations • 8.EE.A.2

Find $\sqrt{9}$ and $\sqrt[3]{(-8)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q1/5

Find $\sqrt[3]{27}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(-4)}$ is a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q3/5

$\sqrt[3]{(-1)}$ equals

A. i

B. undefined

C. -1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q4/5

Find $\sqrt{16} - 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q5/5

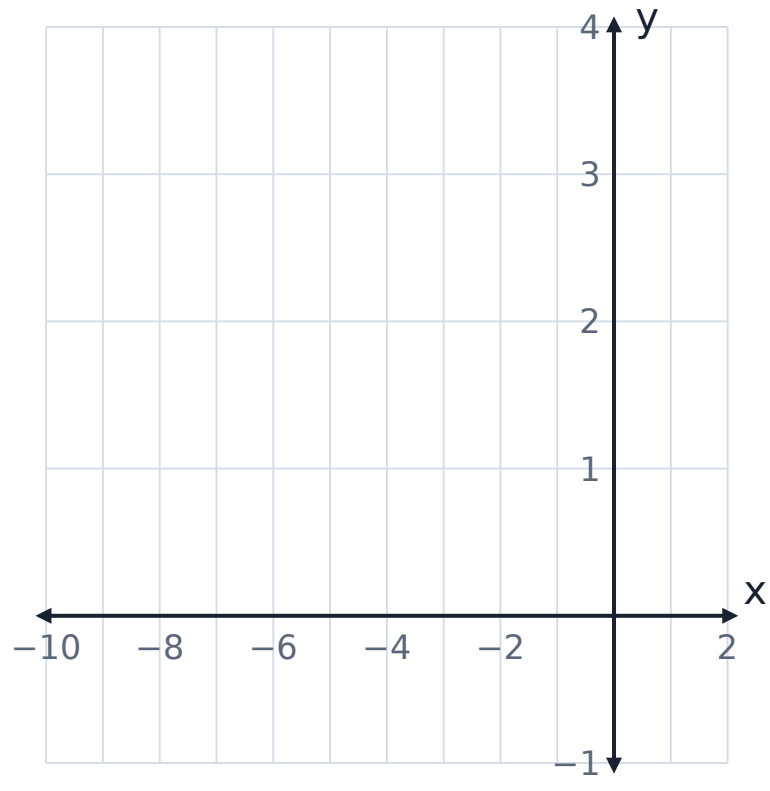
SET 46 • HSF-IF.C.7.b

FORWARD 1

Graph radical functions with horizontal stretches and reflections • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph the function on the grid. Show and label the starting point and two more points, and state the domain.

$y = \sqrt{(-x)}$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q1/5

Show that $\sqrt{(4x)} = 2\sqrt{x}$ for $x \geq 0$ and explain what this says about the two graphs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q2/5

True or false?

$y = \sqrt[3]{(-x)}$ is the same as $y = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q3/5

The domain of $y = \sqrt{(3 - x)}$ is

A. all reals

B. $x \geq 3$

C. $x \leq 3$

D. $x \geq -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q4/5

Write a square root function whose graph starts at $(4, -2)$ and goes up to the left.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSF-IF.C.7.b

FORWARD 2

Graph a radical function and its inverse • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

Find the inverse of $f(x) = \sqrt{(x - 2)} + 1$ and give its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q1/5

Find the inverse of $g(x) = \sqrt[3]{(x + 1)}$.

$g^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q2/5

True or false?

The graph of an inverse is the reflection of the graph over y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q3/5

The point (6, 3) on $f(x) = \sqrt{(x - 2)} + 1$ gives which point on f^{-1}?

A. (6, 3)

B. (3, 6)

C. (−6, −3)

D. (3, −6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q4/5

Explain why the inverse of $f(x) = \sqrt{(x - 2)} + 1$ needs the domain $x \geq 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q5/5

SET 47 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph piecewise-defined and absolute value functions

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate and graph simple piecewise functions	**HSF-IF.C.7.b**
BACKWARD 2	Absolute value as distance	**6.NS.C.7**
BACKWARD 3	Open and closed endpoints	**6.EE.B.8**
FORWARD 1	Write piecewise functions from graphs and contexts	**HSF-IF.C.7.b**
FORWARD 2	Describe absolute value graphs as transformations, and identify…	**HSF-BF.B.3**

SET 47 • HSF-IF.C.7.b

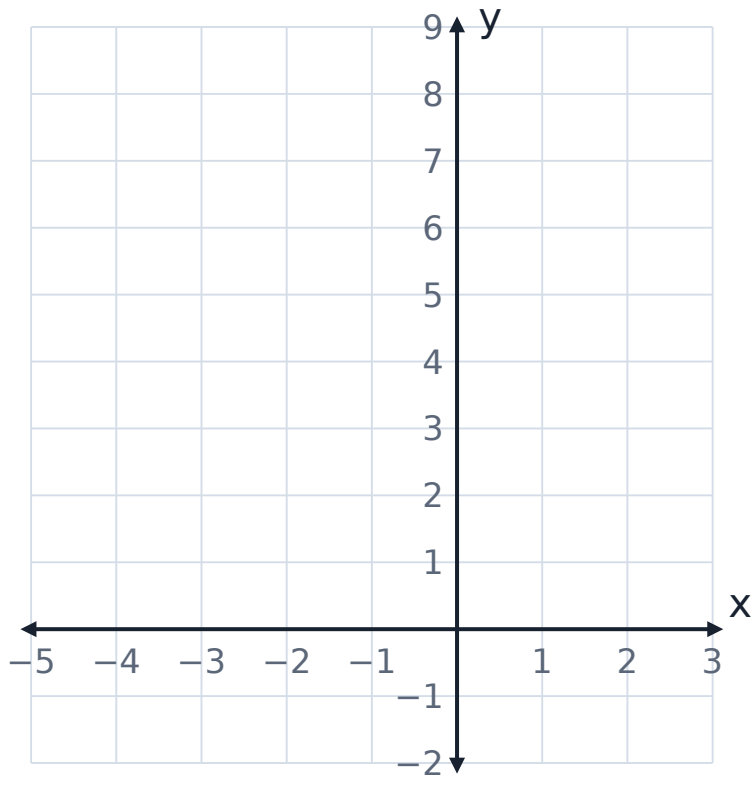
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph the function on the grid. Show and label each endpoint with an open or closed circle.

$f(x) = x + 3$ for $x < -1$, and

$f(x) = x^2$ for $x \geq -1$



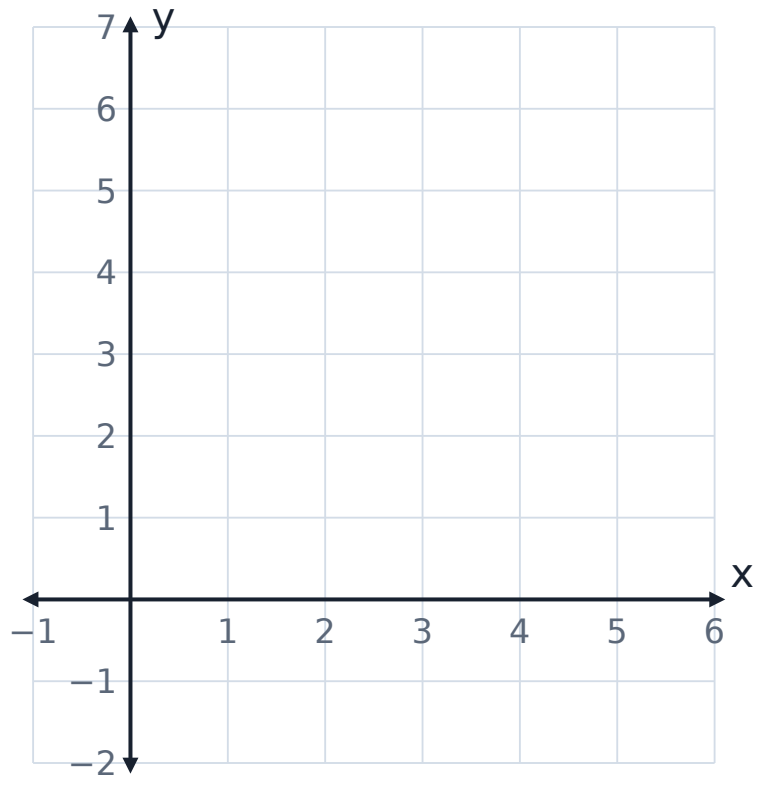
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q1/5

Graph the function on the grid. Show and label the vertex and two more points.

$y = |2x - 4| - 1$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

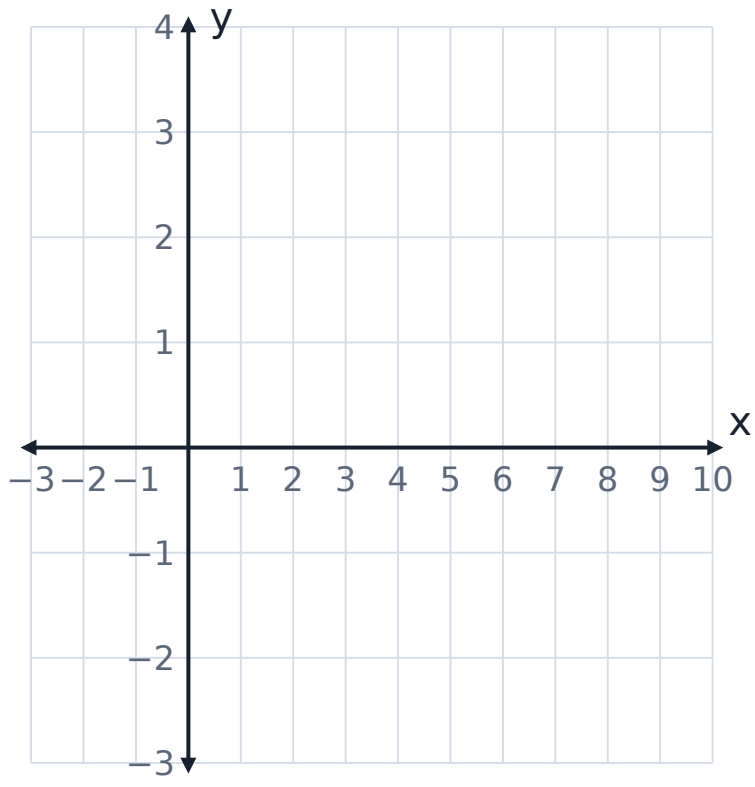
ID S47-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q2/5

Graph the function on the grid. Show and label each endpoint with an open or closed circle.

$f(x) = -2$ for $x \leq 1$, and

$f(x) = \sqrt{(x - 1)}$ for $x > 1$



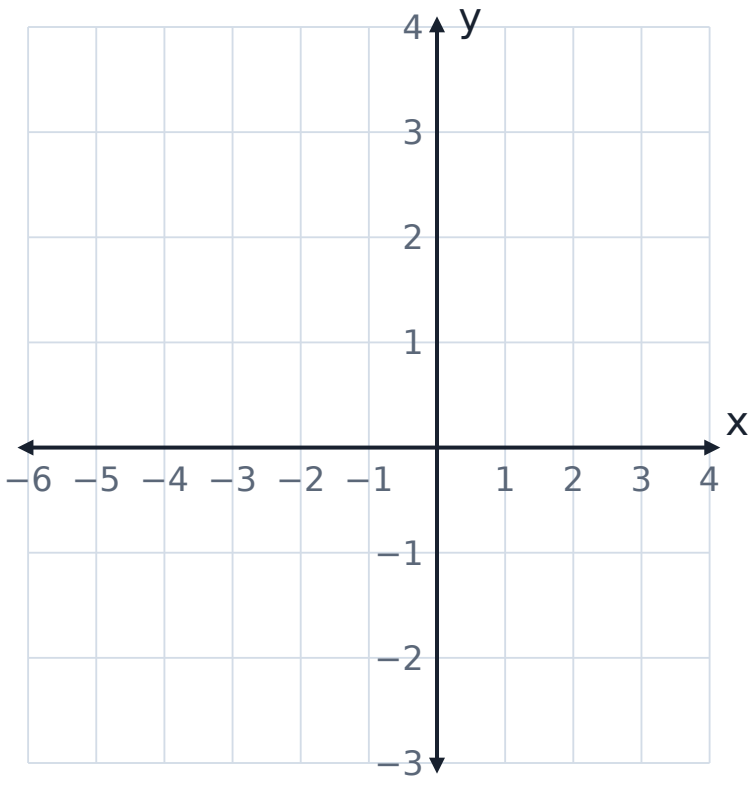
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q3/5

Graph the function on the grid. Show and label the vertex and two more points.

$y = -|x + 1| + 3$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

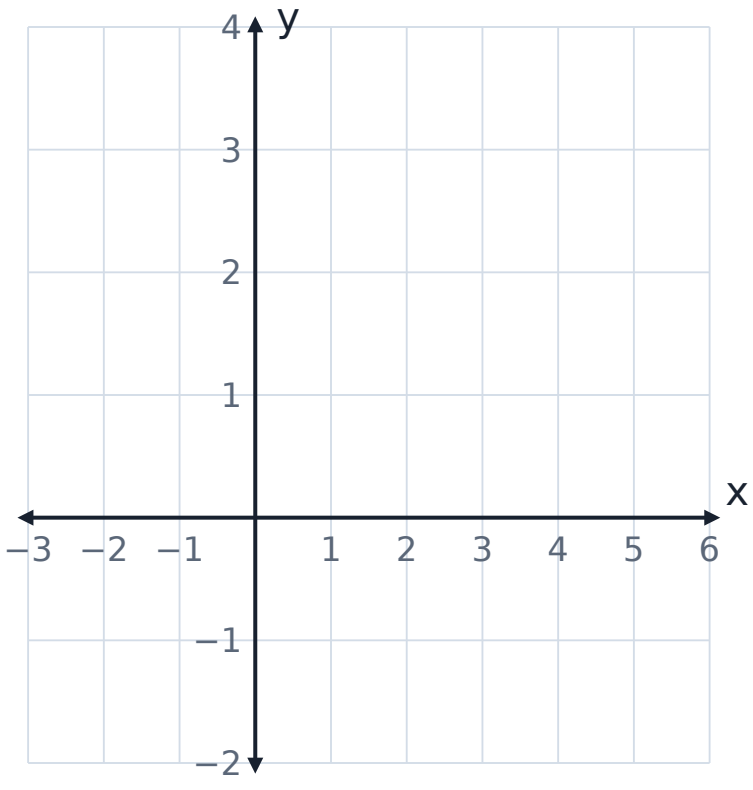
ID S47-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q4/5

Graph the function on the grid. Show and label each endpoint with an open or closed circle.

$f(x) = 2^x$ for $x < 1$, and

$f(x) = 4 - x$ for $x \geq 1$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 1

Evaluate and graph simple piecewise functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

$f(x) = 2$ for $x < 0$ and $f(x) = x$ for $x \geq 0$. Find $f(-3)$ and $f(5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q1/5

$f(x) = 2$ for $x < 0$ and $f(x) = x$ for $x \geq 0$. Find $f(0)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q2/5

True or false?

A piecewise function uses different rules on different parts of its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q3/5

For $f(x) = x + 3$ $(x < -1)$, x^2 $(x \geq -1)$, $f(-1)$ is

A. 3

B. 2

C. -1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q4/5

$f(x) = x + 3$ for $x < -1$, and $f(x) = x^2$ for $x \geq -1$. Find $f(-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 2

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7

Find $|-5|$ and $|3|$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q1/5

Find $|2(1) - 4| - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q2/5

True or false?

$|x|$ can be negative for negative x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q3/5

$|x + 1| = 0$ when

A. $x = 0$

B. $x = -1$

C. $x = 1$

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q4/5

Find $-|0 + 1| + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 3

Open and closed endpoints • Grade 6 • Expressions & Equations • 6.EE.B.8

On a number line, is $x < -1$ drawn with an open or a closed circle at -1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q1/5

Is $x \geq 1$ drawn with an open or a closed circle at 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q2/5

True or false?

An open circle means the endpoint is included.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q3/5

On a number line, $x \leq 1$ is drawn with

A. an open circle and shading right

B. a closed circle at 1 and shading right

C. an open circle at 1 and shading left

D. a closed circle at 1 and shading left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q4/5

Write the inequality for "all numbers greater than 1".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSF-IF.C.7.b

FORWARD 1

Write piecewise functions from graphs and contexts • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

A shipping company charges $5 for packages up to 2 lb and $5 plus $1.50 per pound over 2 lb. Write the cost C(w).

C(w) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q1/5

Write $|x - 3|$ as a piecewise function without absolute value bars.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q2/5

True or false?

A piecewise function can have a jump and still be a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q3/5

Which piecewise function is continuous at $x = 1$?

A. 2^x for $x < 1$, $4 - x$ for $x \geq 1$

B. x^2 for $x < 1$, $2x - 1$ for $x \geq 1$

C. 2 for $x < 1$, 3 for $x \geq 1$

D. x for $x < 1$, $x + 1$ for $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q4/5

Find k so that $f(x) = kx$ for $x < 2$ and $f(x) = x^2$ for $x \geq 2$ has no jump at 2.

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSF-IF.C.7.b

FORWARD 2

Describe absolute value graphs as transformations, and identify symmetry • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Describe $y = |2x - 4| - 1$ as a transformation of $y = |x|$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q1/5

Write an absolute value function with vertex $(-1, 3)$ that opens down and passes through $(2, 0)$.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q2/5

True or false?

$y = |x|$ is an odd function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q3/5

$y = |x - 2|$ is symmetric about

A. the origin

B. $y = 2$

C. $x = 0$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q4/5

Is $y = -|x + 1| + 3$ an even function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q5/5

SET 48 • HSF-IF.C.7.c • INTERPRETING FUNCTIONS

Graph polynomial functions showing zeros and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.c**
BACKWARD 1	Factor out common factors and special products	**HSA-SSE.A.2**
BACKWARD 2	Crossing and touching from multiplicity	**HSA-APR.B.3**
BACKWARD 3	Evaluate polynomials at test points	**HSF-IF.A.2**
FORWARD 1	Locate relative extrema of polynomial graphs	**HSF-IF.C.7.c**
FORWARD 2	Interpret polynomial graphs in context	**HSF-IF.B.4**

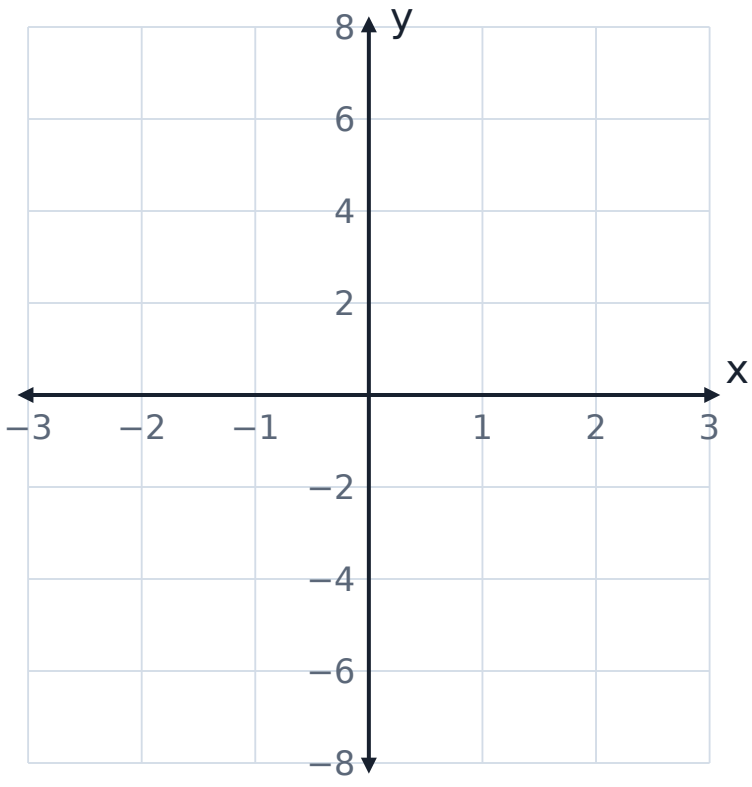
SET 48 • HSF-IF.C.7.c

MAIN

Algebra II main question • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Factor, then graph on the grid. Show every zero, the y-intercept, one point between each pair of zeros, and the end behavior.

$y = x^3 - 4x$



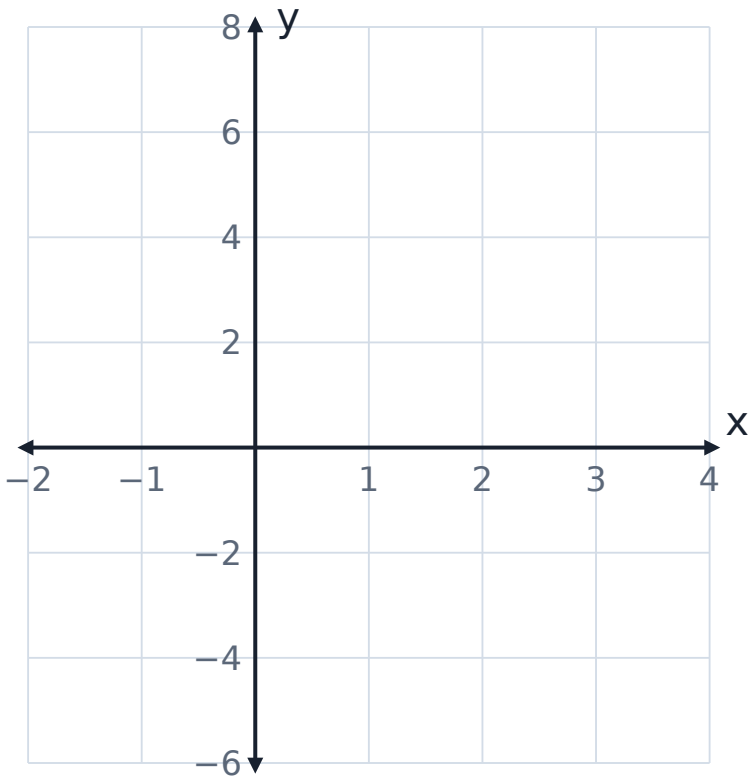
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q1/5

Factor, then graph on the grid. Show every zero, the y-intercept, one point between each pair of zeros, and the end behavior.

$y = -x^3 + 3x^2$



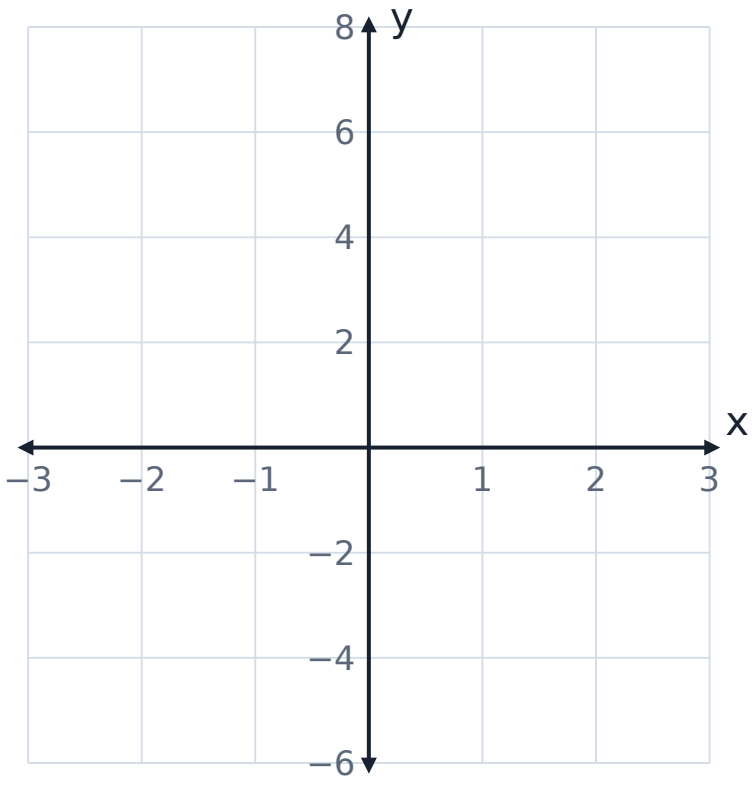
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q2/5

Factor, then graph on the grid. Show every zero, the y-intercept, one point between each pair of zeros, and the end behavior.

$y = x^4 - 4x^2$



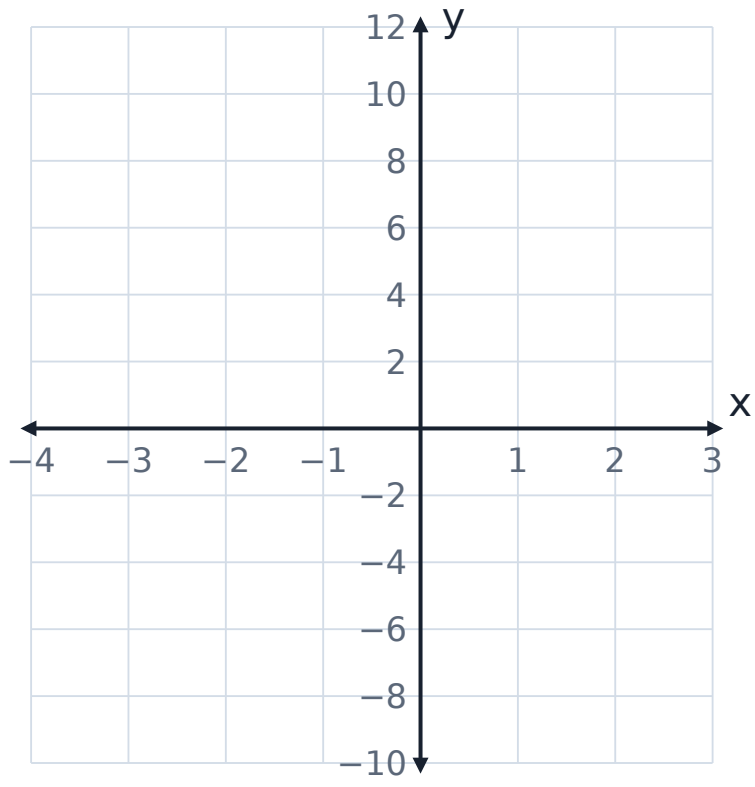
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q3/5

Factor, then graph on the grid. Show every zero, the y-intercept, one point between each pair of zeros, and the end behavior.

$y = x^3 + x^2 - 6x$



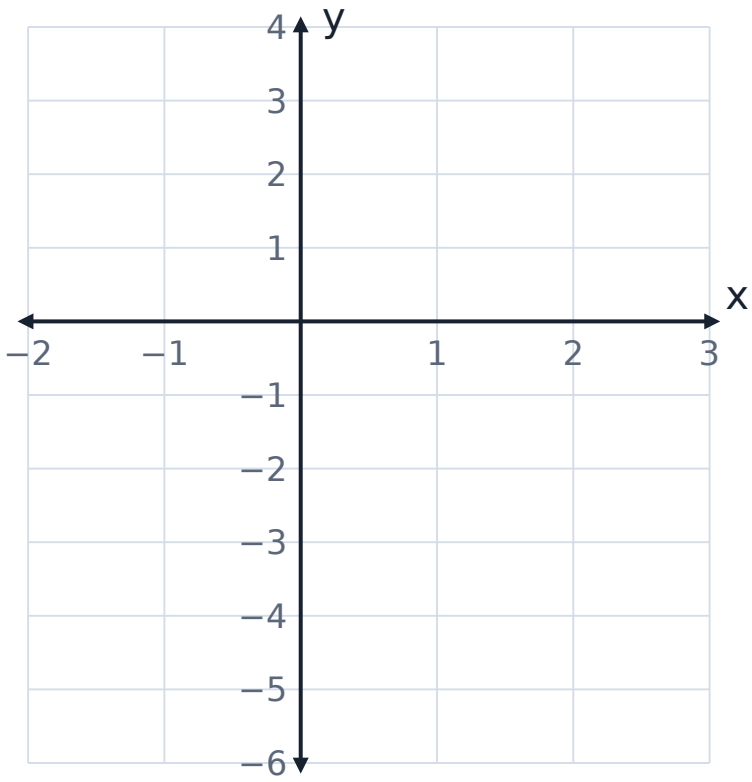
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q4/5

Factor, then graph on the grid. Show every zero, the y-intercept, one point between each pair of zeros, and the end behavior.

$y = -x^4 + 2x^3$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 1

Factor out common factors and special products • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q1/5

Factor $-x^3 + 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q2/5

True or false?

$$x^4 - 4x^2 = x^2(x^2 - 4) = x^2(x - 2)(x + 2).$$

A. True

B. False

$x^3 + x^2 - 6x$ equals

A. $(x + 3)(x - 2)$

B. $x(x + 3)(x - 2)$

C. $x(x - 3)(x + 2)$

D. $x(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q4/5

Factor $-x^4 + 2x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 2

Crossing and touching from multiplicity • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Does $y = x^2(x - 3)$ cross or touch the x-axis at 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q1/5

Does $y = x^3(x - 2)$ cross at 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q2/5

True or false?

A zero of odd multiplicity is a crossing point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q3/5

The graph of $y = (x + 2)^2(x - 1)$ at $x = -2$

A. crosses

B. touches

C. has an asymptote

D. is undefined

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q4/5

Give the end behavior of $y = -x^4 + 2x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 3

Evaluate polynomials at test points • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Evaluate $x^3 - 4x$ at $x = -1$ and $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q1/5

Evaluate $-x^3 + 3x^2$ at $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q2/5

True or false?

$x^4 - 4x^2$ at $x = 1$ equals 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q3/5

$x^3 + x^2 - 6x$ at $x = -2$ equals

A. 8

B. 0

C. -8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q4/5

Evaluate $-x^4 + 2x^3$ at $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • HSF-IF.C.7.c

FORWARD 1

Locate relative extrema of polynomial graphs • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Use a table with steps of 0.1 (or technology) to estimate the relative maximum of $y = x^3 - 4x$ to the nearest tenth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q1/5

Use technology or a table to estimate the relative minimum of $y = x^3 + x^2 - 6x$ on $0 < x < 2$ to the nearest tenth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 4 always has exactly 3 turning points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q3/5

How many turning points can $y = x^4 - 4x^2$ have?

A. 1

B. 4

C. 3

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q4/5

Explain why the graph of $y = x^3 - 4x$ has a turning point between 0 and 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSF-IF.C.7.c

FORWARD 2

Interpret polynomial graphs in context • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

A box's volume is $V(x) = x(12 - 2x)^2$. Which part of the polynomial graph is meaningful, and what is the maximum volume there?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q1/5

A company's profit (thousands of dollars) is $P(x) = -x^3 + 9x^2 - 15x$ for x hundred items, $0 \le x \le 8$. When is the profit positive?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q2/5

True or false?

A zero of a profit model is a break-even point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q3/5

A population model $P(t) = -t^3 + 6t^2 + 15t + 100$ rises then falls. The relative maximum means

A. the largest population before decline

B. the starting population

C. the end behavior

D. the time the population is 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q4/5

Why does the end behavior of $V(x) = x(12 - 2x)^2$ not matter for the box?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q5/5

SET 49 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph exponential functions with intercepts, asymptote, and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Graph basic exponential functions	**HSF-IF.C.7.e**
BACKWARD 2	Zero and negative exponents	**8.EE.A.1**
BACKWARD 3	Vertical and horizontal shifts and reflections	**HSF-BF.B.3**
FORWARD 1	Compare and describe exponential graphs	**HSF-IF.C.7.e**
FORWARD 2	Graph the inverse of an exponential function	**HSF-BF.B.4.a**

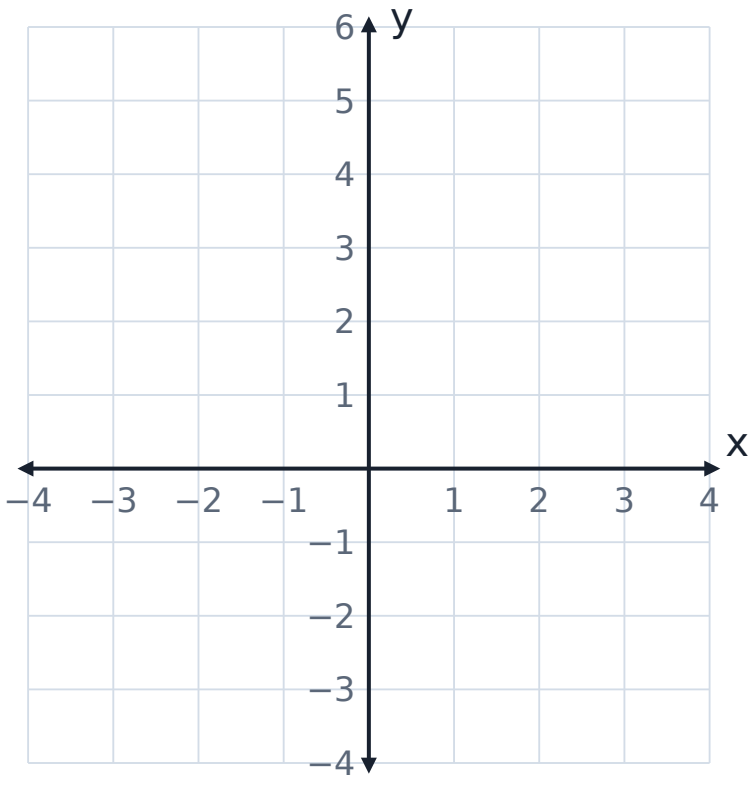
SET 49 • HSF-IF.C.7.e

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph the function on the grid. Show and label the y-intercept, the x-intercept (as a decimal), and the asymptote (dashed).

$y = 2^x - 3$



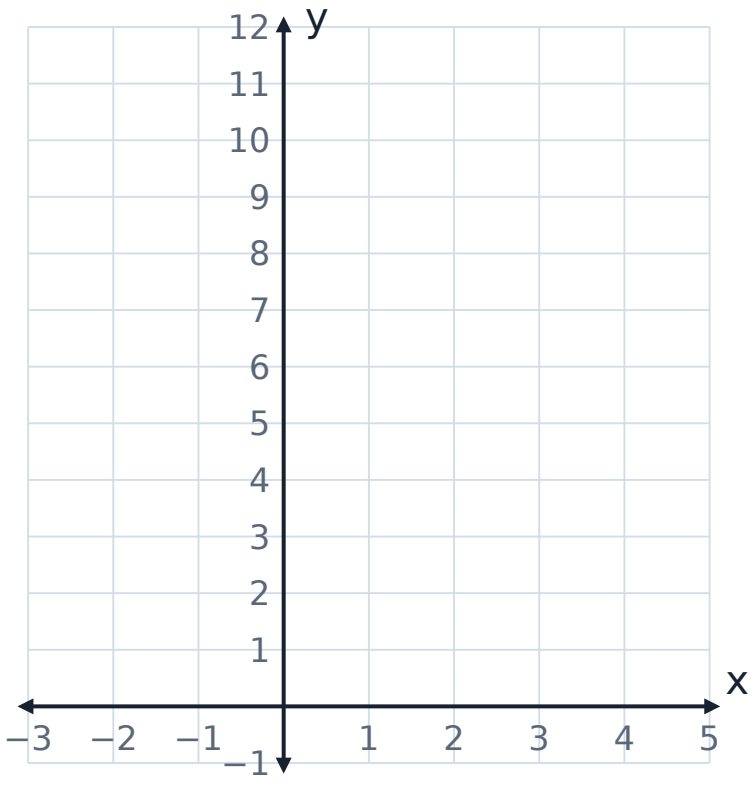
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q1/5

Graph the function on the grid. Show and label the y-intercept and the asymptote (dashed), and describe the end behavior.

$y = 3(0.5)^x$



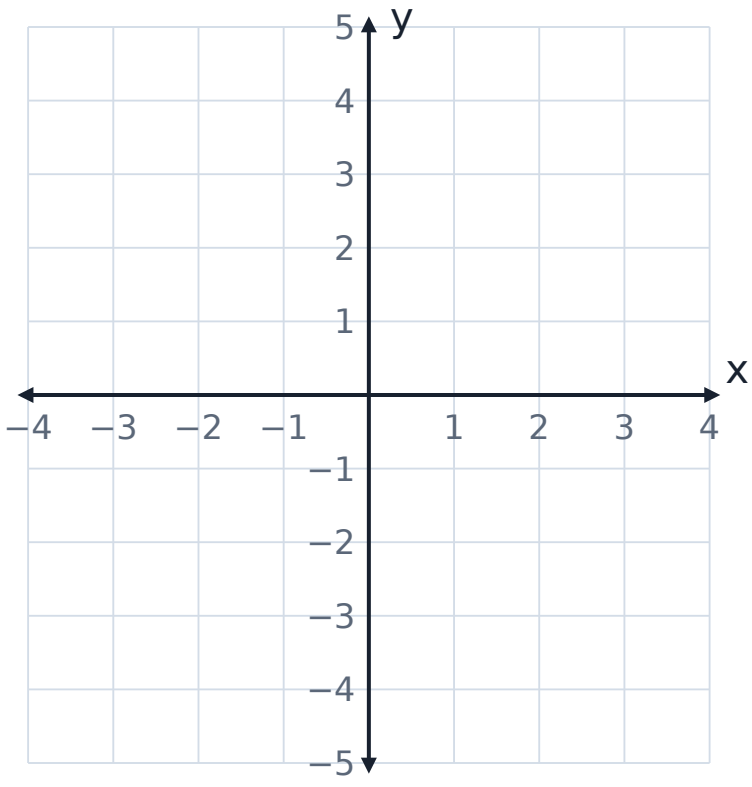
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q2/5

Graph the function on the grid. Show and label the y-intercept, the x-intercept, and the asymptote (dashed).

$y = -2^x + 4$



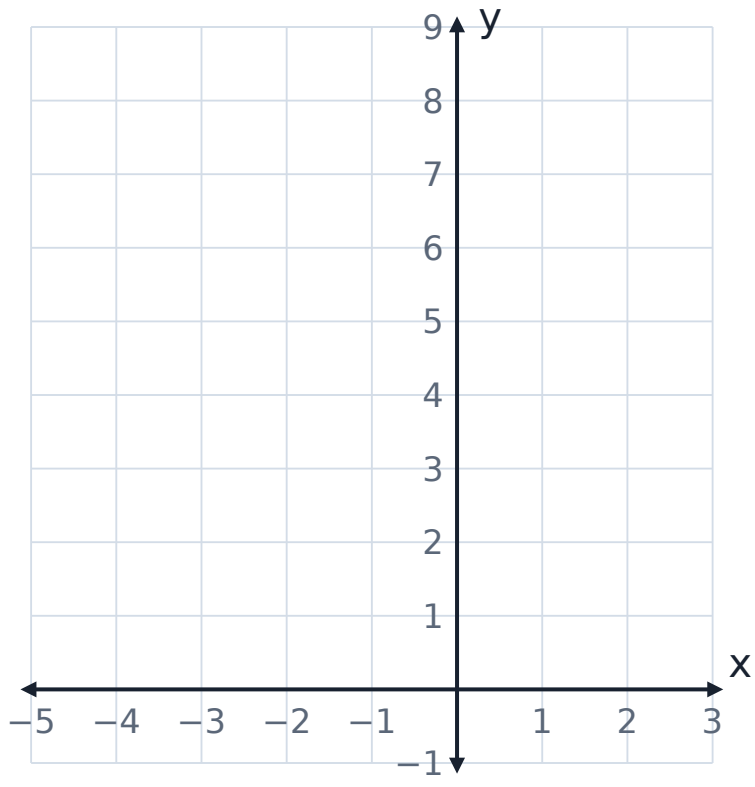
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q3/5

Graph the function on the grid. Show and label the y-intercept and the asymptote (dashed).

$y = 2^{x+1}$



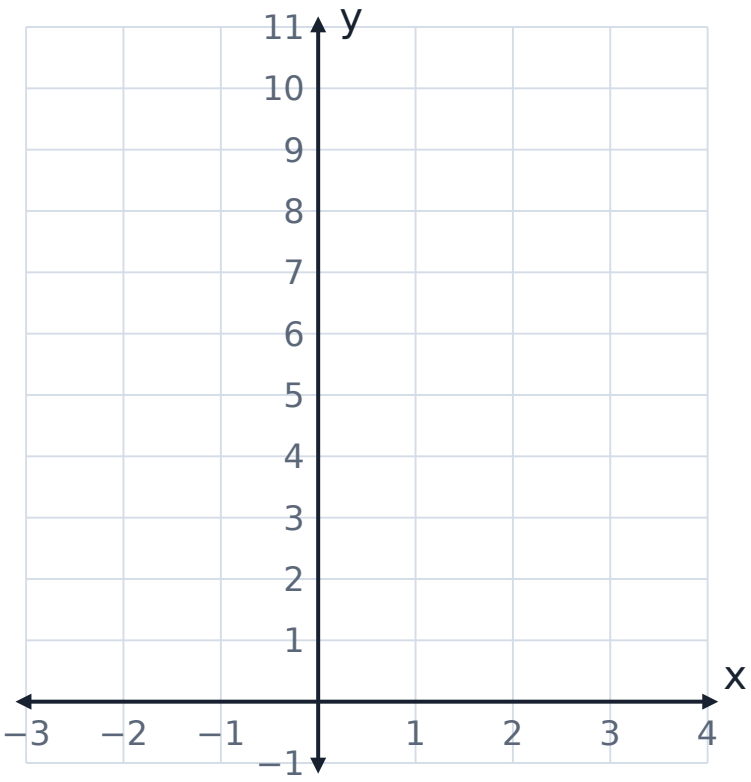
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q4/5

Graph the function on the grid. Show and label the y-intercept and the asymptote (dashed).

$y = (1/3)^x + 1$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 1

Graph basic exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Complete the table for $y = 2^x$ at $x = -2, -1, 0, 1, 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q1/5

What is the y-intercept of $y = 3(0.5)^x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q2/5

True or false?

$y = 2^x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q3/5

$y = (0.5)^x$ is

A. linear

B. increasing

C. constant

D. decreasing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q4/5

Does $y = 2^x$ have an x-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 2

Zero and negative exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Evaluate 2^{-2}.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q1/5

Evaluate $(1/3)^{-2}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q2/5

True or false?

$b^0 = 0$ for $b \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q3/5

$3(0.5)^{-1}$ equals

A. 6

B. 3

C. -1.5

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q4/5

Evaluate $2^{-1} - 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 3

Vertical and horizontal shifts and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How does $y = 2^x - 3$ relate to $y = 2^x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q1/5

How does $y = 2^{x+1}$ relate to $y = 2^{x}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q2/5

True or false?

A vertical shift moves the horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q3/5

$y = -2^x + 4$ has asymptote

A. $y = 4$

B. $y = 0$

C. $y = -4$

D. $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q4/5

Give the asymptote of $y = (1/3)^x + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • HSF-IF.C.7.e

FORWARD 1

Compare and describe exponential graphs • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Explain why the graphs of $y = 2^x$ and $y = (1/2)^x$ are reflections of each other over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q1/5

Show that $2^{x+1} = 2 \cdot 2^{x}$ and explain what this says about two transformations.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q2/5

True or false?

$y = 5 \cdot 3^x$ and $y = 3^x$ have different horizontal asymptotes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q3/5

Which exponential graph passes through (0, 4) and (1, 2)?

A. $y = 2^x + 3$

B. $y = 4 \cdot 2^x$

C. $y = 0.5 \cdot 4^x$

D. $y = 4(0.5)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q4/5

Write an exponential function with asymptote $y = -1$ and y-intercept 2 that is increasing.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSF-IF.C.7.e

FORWARD 2

Graph the inverse of an exponential function • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

Write the inverse of $f(x) = 2^x$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q1/5

The point (3, 8) is on $y = 2^x$. Which point is on its inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q2/5

True or false?

The inverse of $y = 2^x$ has the vertical asymptote $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q3/5

The inverse of $y = 2^x - 3$ is

A. $y = \log_2(x - 3)$

B. $y = \log_2 x - 3$

C. $y = 2^{x+3}$

D. $y = \log_2(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q4/5

Give the domain and range of $y = \log_2 x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q5/5

SET 50 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph logarithmic functions with intercept, asymptote, and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Graph exponential functions	**HSF-IF.C.7.e**
BACKWARD 2	Find inverses by swapping inputs and outputs	**HSF-BF.B.4.a**
BACKWARD 3	Shift and reflect graphs	**HSF-BF.B.3**
FORWARD 1	Graph logarithms with reflections that change the domain	**HSF-IF.C.7.e**
FORWARD 2	Solve equations with logarithms	**HSF-LE.A.4**

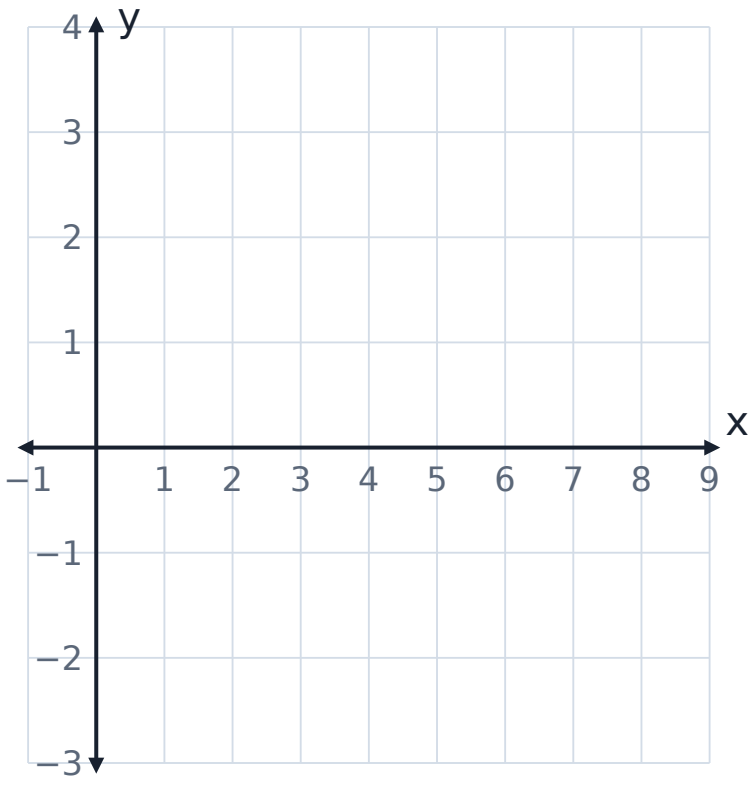
SET 50 • HSF-IF.C.7.e

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph the function on the grid. Show and label the x-intercept, two more points, and the asymptote (dashed); state the domain.

$y = \log_2 x$



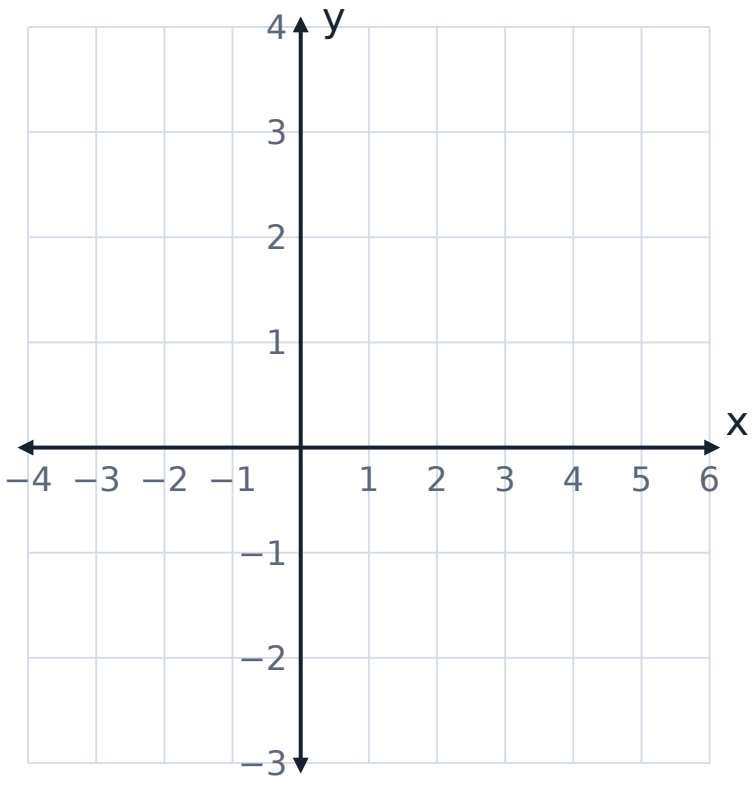
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q1/5

Graph the function on the grid. Show and label the x-intercept, two more points, and the asymptote (dashed); state the domain.

$y = \log_2(x + 3)$



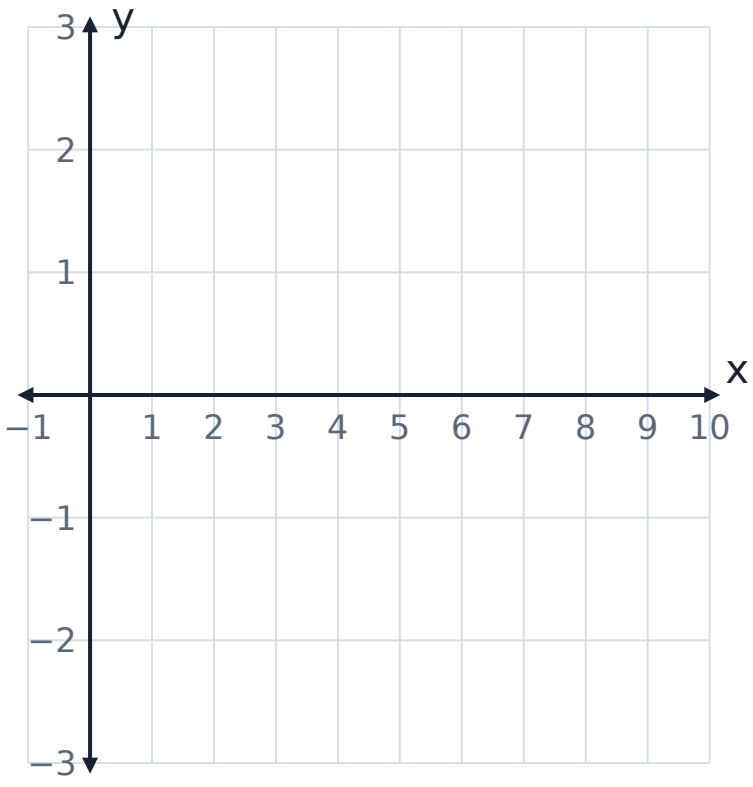
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q2/5

Graph the function on the grid. Show and label the x-intercept, two more points, and the asymptote (dashed); state the domain.

$y = \log_3 x - 1$



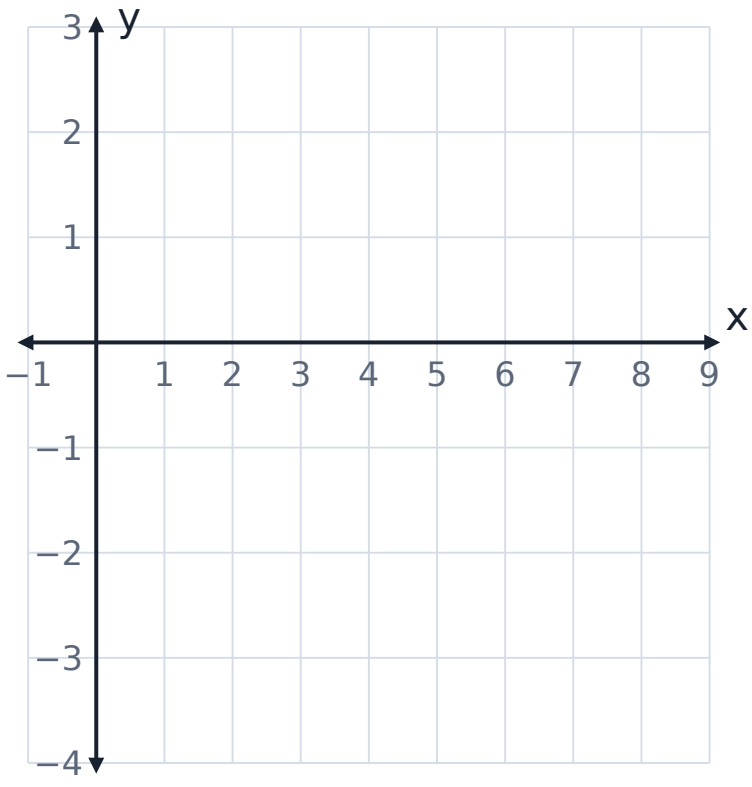
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q3/5

Graph the function on the grid. Show and label the x-intercept, two more points, and the asymptote (dashed); state the domain.

$y = -\log_2 x$



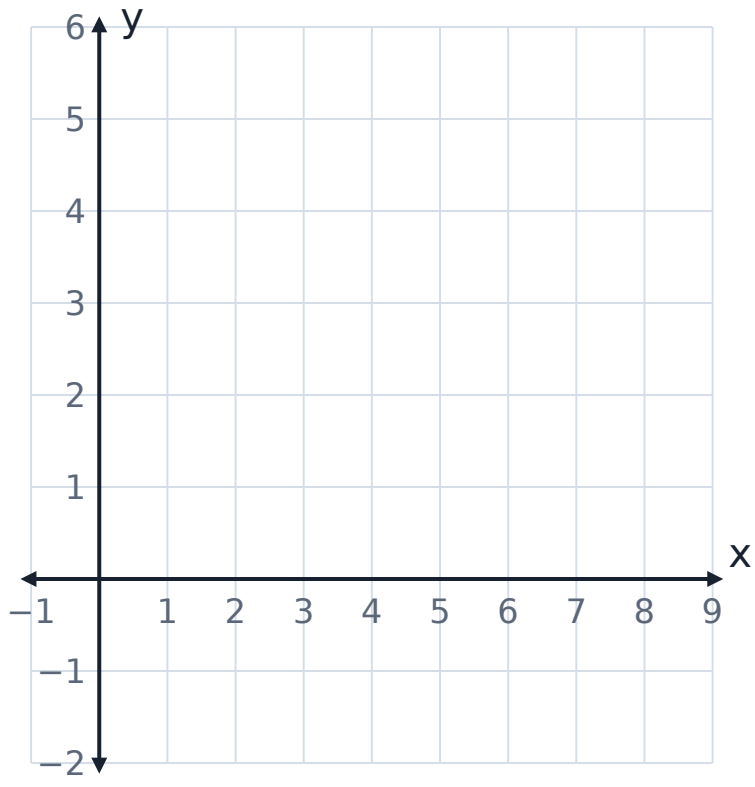
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q4/5

Graph the function on the grid. Show and label two points and the asymptote (dashed); state the domain and whether there is an x-intercept.

$$y = \log_2(x - 1) + 2$$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 1

Graph exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Give three points on $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q1/5

Give the asymptote of $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q2/5

True or false?

$y = 3^x$ passes through (3, 1).

A. True

B. False

$y = 2^x$ passes through

A. (2, 1)

B. (−1, −2)

C. (1/2, −1)

D. (−1, 1/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q4/5

Give the range of $y = 3^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 2

Find inverses by swapping inputs and outputs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

(2, 4) is on $y = 2^x$. Which point is on its inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

Write the inverse of $y = 3^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

True or false?

Swapping x and y in $y = 2^x$ gives $x = 2^y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

Which point is on $y = \log_2 x$?

A. (3, 8)

B. (2, 4)

C. (8, 3)

D. (0, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

What is $\log_2 16$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 3

Shift and reflect graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How does $y = \log_2(x + 3)$ relate to $y = \log_2 x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q1/5

How does $y = -\log_2 x$ relate to $y = \log_2 x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q2/5

True or false?

A horizontal shift moves the vertical asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q3/5

$y = \log_2(x - 1) + 2$ has asymptote

A. $x = -1$

B. $y = 2$

C. $x = 1$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q4/5

Where does (1, 0) of $y = \log_3 x$ move under $y = \log_3 x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q5/5

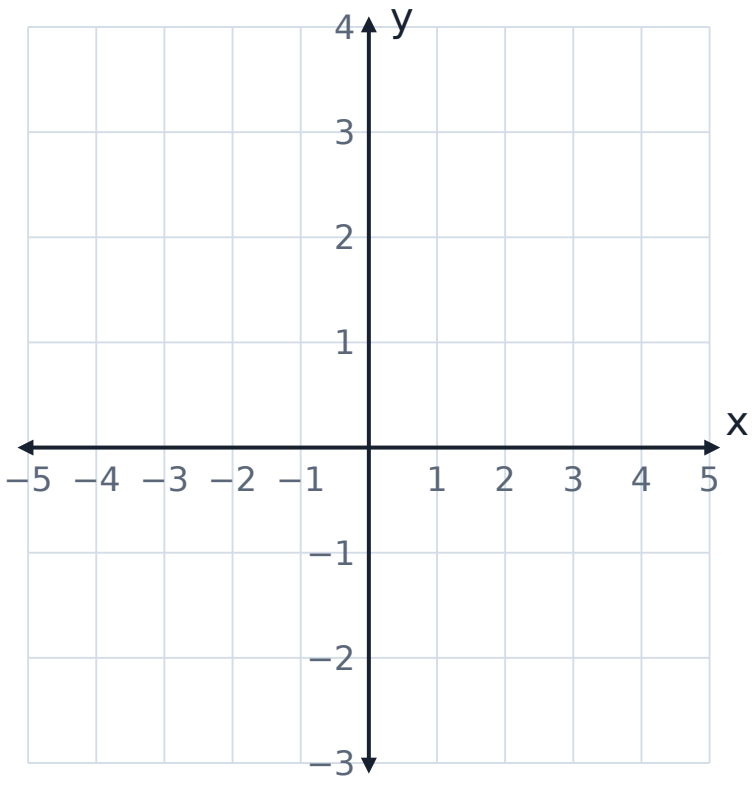
SET 50 • HSF-IF.C.7.e

FORWARD 1

Graph logarithms with reflections that change the domain • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph the function on the grid. Show and label the x-intercept, two more points, and the asymptote (dashed); state the domain.

$y = \log_2(4 - x)$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q1/5

Give the domain of $y = \log(x^2 - 1)$ and explain the graph has two branches.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q2/5

True or false?

$y = \log_2(2x)$ is $y = \log_2 x$ shifted up 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q3/5

The x-intercept of $y = \log_2(x + 3)$ is

A. -2

B. 1

C. -3

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q4/5

Find the x-intercept of $y = \log_2(x - 1) + 2$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSF-IF.C.7.e

FORWARD 2

Solve equations with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $2^x = 12$ exactly and to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

Solve $5 \cdot 10^{x} = 40$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

True or false?

$\log_2 12$ is between 4 and 5.

A. True

B. False

The solution of $3^x = 7$ is

A. log 7/log 3

B. log 3/log 7

C. 7/3

D. log 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

Use the graph of $y = \log_2 x$ to estimate $\log_2 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

SET 51 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph trigonometric functions showing period, midline, and amplitude

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Unit-circle values at quarter turns	**HSF-TF.A.2**
BACKWARD 2	Read amplitude, period, and midline from a formula	**HSF-TF.B.5**
BACKWARD 3	Vertical stretches, shifts, and reflections	**HSF-BF.B.3**
FORWARD 1	Graph shifted sinusoids and the tangent function	**HSF-IF.C.7.e**
FORWARD 2	Horizontal stretches f(kx) of trigonometric graphs	**HSF-BF.B.3**

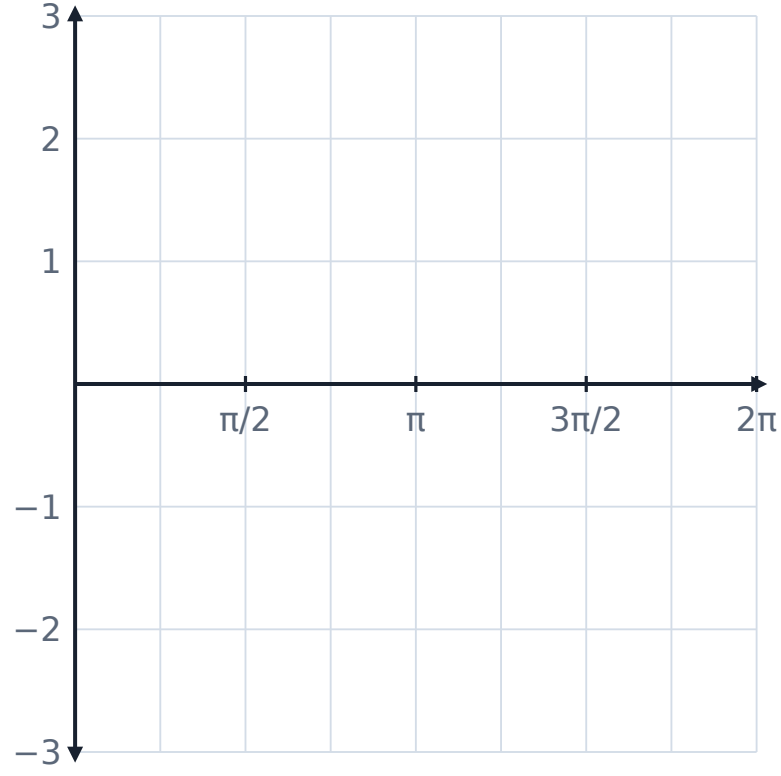
SET 51 • HSF-IF.C.7.e

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph the function on the grid. Show and label the midline (dashed), the maximum and minimum points, and the period for one cycle on $0 \leq x \leq 2\pi$.

$y = 2\sin x$



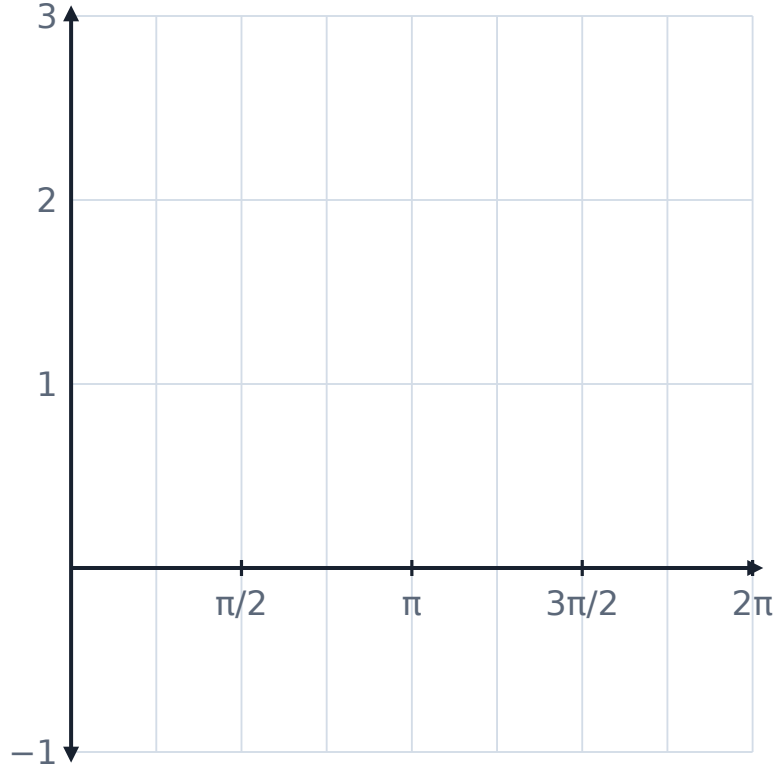
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q1/5

Graph the function on the grid. Show and label the midline (dashed), the maximum and minimum points, and the period on $0 \leq x \leq 2\pi$.

$y = \cos x + 1$



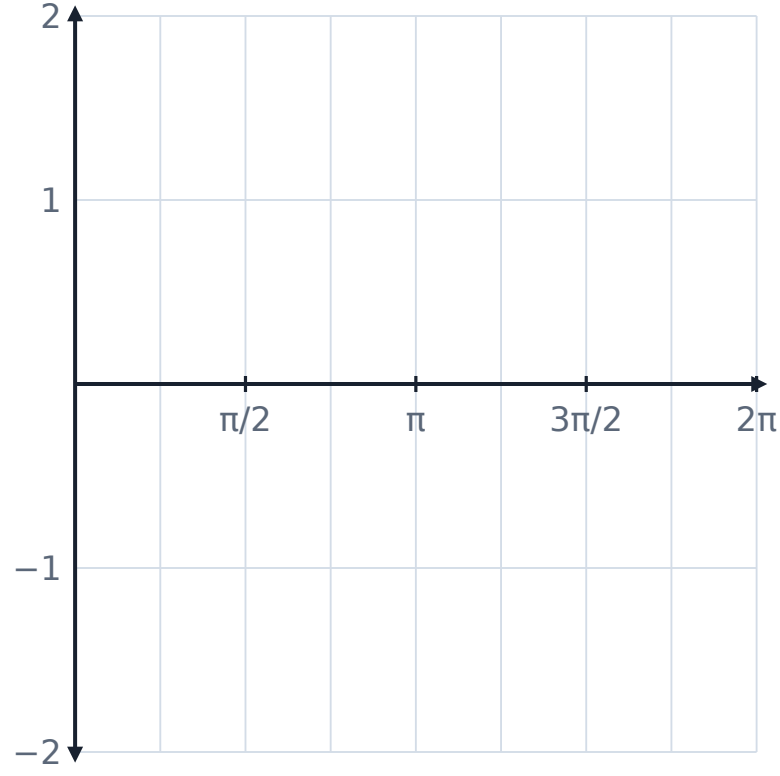
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q2/5

Graph the function on the grid. Show and label the midline (dashed), the maximum and minimum points, and the period on $0 \leq x \leq 2\pi$.

$y = \sin(2x)$



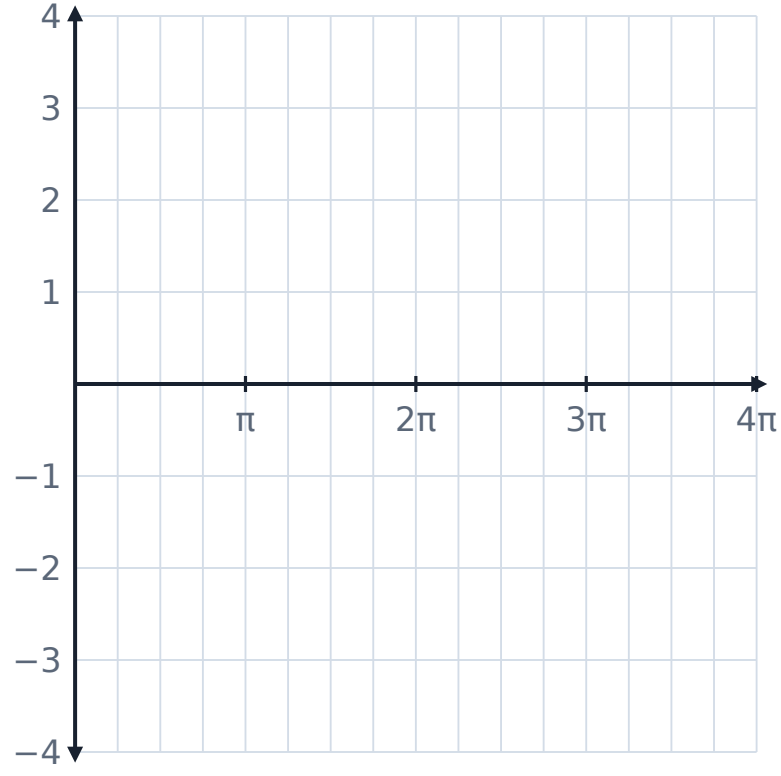
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q3/5

Graph the function on the grid. Show and label the midline (dashed), the maximum and minimum points, and the period on $0 \leq x \leq 4\pi$.

$y = 3\cos(x/2)$



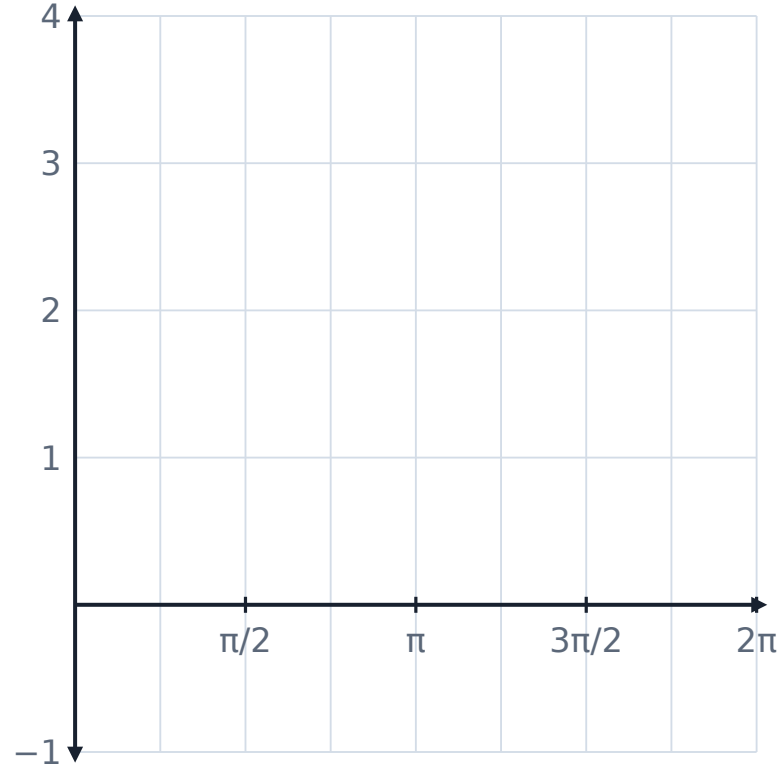
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q4/5

Graph the function on the grid. Show and label the midline (dashed), the maximum and minimum points, and the period on $0 \leq x \leq 2\pi$.

$y = -\sin x + 2$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 1

Unit-circle values at quarter turns • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Give sin x at x = 0, π/2, π, 3π/2, 2π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q1/5

Give cos x at x = 0, π/2, π, 3π/2, 2π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q2/5

True or false?

$\sin(3\pi/2) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q3/5

cos(π) equals

A. π

B. 1

C. −1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q4/5

Give $\sin(\pi/4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 2

Read amplitude, period, and midline from a formula • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Give the amplitude and period of $y = 3\cos(x/2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q1/5

Give the midline of $y = -\sin x + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q2/5

True or false?

$y = \sin(2x)$ has period 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q3/5

$y = 2\sin x$ has amplitude

A. 1

B. 2

C. π

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q4/5

Give the period of $y = \cos(3x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 3

Vertical stretches, shifts, and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

Describe $y = \cos x + 1$ compared with $y = \cos x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q1/5

Describe $y = -\sin x$ compared with $y = \sin x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q2/5

True or false?

$y = 2\sin x$ has the same zeros as $y = \sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q3/5

The maximum of $y = -\sin x + 2$ is

A. 1

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q4/5

What is the minimum of $y = \cos x + 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q5/5

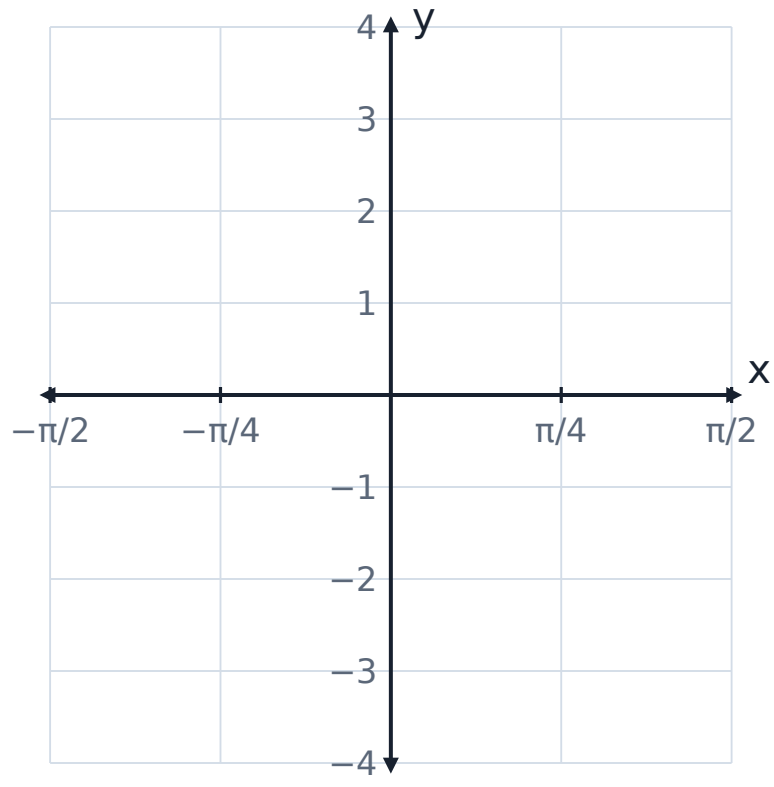
SET 51 • HSF-IF.C.7.e

FORWARD 1

Graph shifted sinusoids and the tangent function • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph the function on the grid. Show and label the asymptotes (dashed) and three points.

$y = \tan x$ for $-\pi/2 < x < \pi/2$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q1/5

Describe the graph of $y = \sin(x - \pi/2)$ compared with $y = \sin x$, and name a simpler formula for it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q2/5

True or false?

The period of $y = \tan x$ is 2π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q3/5

The first maximum of $y = \cos(x - \pi/3)$ for $x \geq 0$ is at

A. $x = 2\pi/3$

B. $x = \pi/3$

C. $x = 0$

D. $x = -\pi/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q4/5

Give the x-values of the maximum points of $y = \sin(x + \pi/4)$ on $0 \leq x < 2\pi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSF-IF.C.7.e

FORWARD 2

Horizontal stretches f(kx) of trigonometric graphs • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

How is $y = \sin(2x)$ obtained from $y = \sin x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q1/5

How is $y = \cos(x/2)$ obtained from $y = \cos x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q2/5

True or false?

Replacing x by kx (k > 1) shortens the period by the factor k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q3/5

$y = \sin(-x)$ is the same as

A. $\cos x$

B. $\sin x$

C. $-\cos x$

D. $-\sin x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q4/5

For which k does $y = \cos(kx)$ have period 6π?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q5/5

SET 52 • HSF-IF.C.8.b • INTERPRETING FUNCTIONS

Rewrite exponential functions to reveal rates over other time units

MAIN	Algebra II main question	**HSF-IF.C.8.b**
BACKWARD 1	Read growth rates from exponential functions	**HSF-IF.C.8.b**
BACKWARD 2	Powers of powers and rational exponents	**HSN-RN.A.1**
BACKWARD 3	Percent change and growth factors	**7.RP.A.3**
FORWARD 1	Compare growth offers by rewriting to the same time unit	**HSF-IF.C.8.b**
FORWARD 2	Find doubling times and half-lives with logarithms	**HSF-LE.A.4**

SET 52 • HSF-IF.C.8.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

$y = 100(1.08)^t$ with t in years. Rewrite it with a monthly growth factor and give the monthly percent rate to the nearest hundredth of a percent. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q1/5

$y = 500(0.97)^t$ with t in years. Rewrite it per decade and give the percent decrease per decade to the nearest hundredth of a percent. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q2/5

$Q = 50(2)^{t/3}$ with t in hours. Rewrite it with an hourly growth factor and give the hourly percent rate to the nearest hundredth of a percent. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q3/5

$A = 1000(1.005)^{12t}$ with t in years. Rewrite it with an annual growth factor and give the annual percent rate to the nearest hundredth of a percent. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q4/5

$N = 800(0.5)^{t/10}$ with t in years. Rewrite it with a yearly factor and give the yearly percent decrease to the nearest hundredth of a percent. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 1

Read growth rates from exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

Is $y = 100(1.08)^t$ growth or decay, and at what percent per year?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q1/5

Is $y = 500(0.97)^t$ growth or decay, and at what percent per year?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q2/5

True or false?

$y = 2(1.25)^t$ grows 125% per unit of t.

A. True

B. False

$y = 40(0.9)^t$ decreases by

A. 9% per unit

B. 10% per unit

C. 90% per unit

D. 0.9% per unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q4/5

What is the growth factor for 6.17% growth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 2

Powers of powers and rational exponents • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

Rewrite 1.08^{t} as a power of $1.08^{1/12}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q1/5

Rewrite $2^{t/3}$ as $(2^{1/3})^{t}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q2/5

True or false?

$(b^m)^n = b^{mn}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q3/5

1.005^{12t} equals

A. $(1.005^{12})^{t}$

B. $(1.005^{t})/12$

C. $12 \cdot 1.005^{t}$

D. $1.005^{12} + t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q4/5

Write $8^{1/3}$ as a whole number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 3

Percent change and growth factors • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Write the growth factor for a 25.99% increase.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q1/5

A factor 0.7374 means what percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q2/5

True or false?

A factor 1.006434 means about 0.64% growth.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q3/5

A 3% decrease corresponds to the factor

A. 0.03

B. 0.97

C. 0.7

D. 1.03

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q4/5

Write the percent change for a factor of 0.933.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • HSF-IF.C.8.b

FORWARD 1

Compare growth offers by rewriting to the same time unit • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Account A grows 0.5% per month. Account B grows 6.1% per year. Which grows faster? Rewrite to compare. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q1/5

A culture doubles every 3 hours; another triples every 5 hours. Which grows faster per hour? A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q2/5

True or false?

Growing 1% per month is the same as growing 12% per year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q3/5

Which is the greatest annual growth?

A. 4% per half year

B. 0.6% per month

C. 8% per year

D. 2% per quarter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q4/5

A half-life of 10 years and a decay of 7% per year: which decays faster? A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSF-IF.C.8.b

FORWARD 2

Find doubling times and half-lives with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Find the doubling time of $y = 100(1.08)^t$, to the nearest hundredth of a year. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q1/5

Find the half-life of $y = 500(0.97)^t$ to the nearest hundredth of a year. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q2/5

True or false?

The doubling time of $y = 50(2)^{t/3}$ is 6 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q3/5

The doubling time of a quantity growing 6.17% per year is about

A. 16.2 years

B. 11.6 years

C. 6.17 years

D. 3.1 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q4/5

Explain why rewriting $y = 100(1.08)^t$ as $100 \cdot 2^{t/9.01}$ shows the doubling time.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q5/5

SET 53 • HSF-IF.C.9 • INTERPRETING FUNCTIONS

Compare properties of Algebra II functions in different representations

MAIN	Algebra II main question	**HSF-IF.C.9**
BACKWARD 1	Compare linear, quadratic, and exponential functions	**HSF-IF.C.9**
BACKWARD 2	Recognize a pattern in a table	**8.F.A.3**
BACKWARD 3	Domains, asymptotes, and periods of Algebra II families	**HSF-IF.B.5**
FORWARD 1	Compare rates of change and long-run behavior across represent…	**HSF-IF.C.9**
FORWARD 2	Build a difference function to compare	**HSF-BF.A.1.b**

SET 53 • HSF-IF.C.9

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = 3(1.1)^x$. The table shows g. Which has the greater y-intercept, and which is greater at x = 20? A calculator may be used.

x	0	1	2	3	4
g(x)	10	15	20	25	30

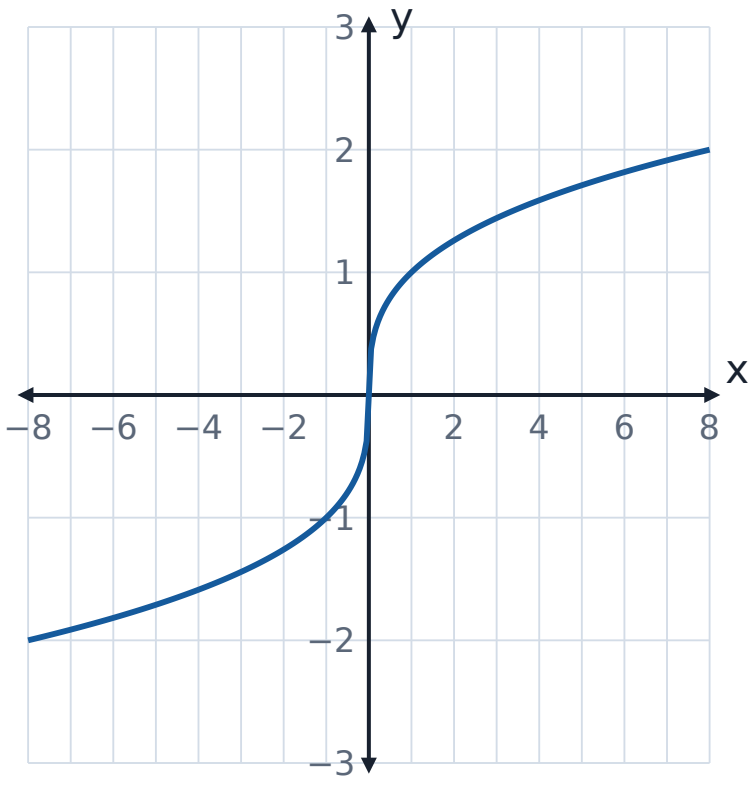
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q1/5

$f(x) = \sqrt{(x - 1)}$. The graph of g is shown. Compare their domains.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q2/5

$h(t) = 12 + 10\sin(\pi t/15)$ gives one Ferris wheel's height (m, t in s). A second wheel has radius 8 m, center 10 m high, and turns once every 40 s. Compare the maximum heights and the periods.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q3/5

$f(x) = \log_2 x$. The table shows g. Which is greater at $x = 9$?

x	1	3	9	27
g(x)	0	1	2	3

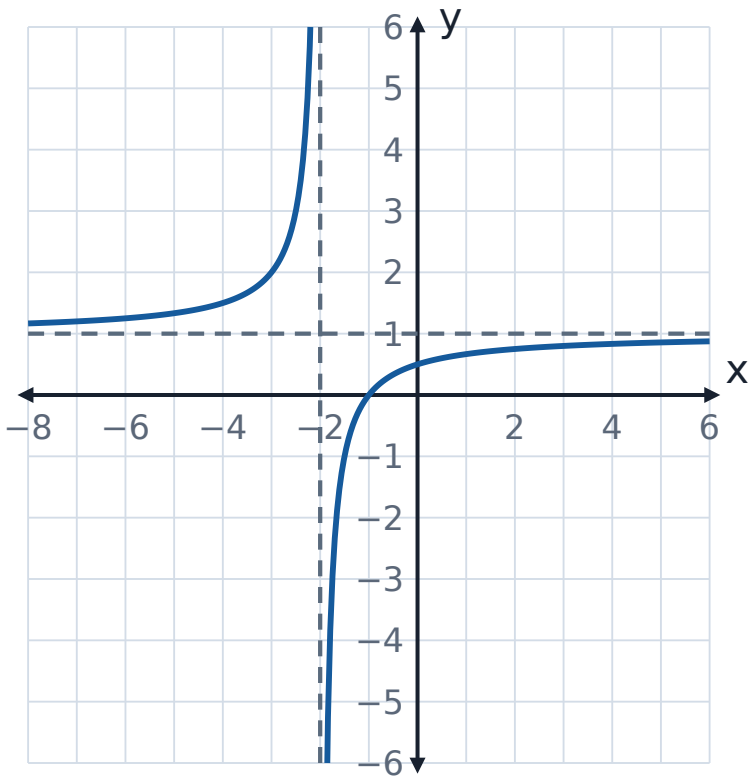
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q4/5

$f(x) = (2x + 1)/(x - 3)$. The graph of g shows the horizontal asymptote $y = 1$ and the vertical asymptote $x = -2$. Compare the asymptotes.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 1

Compare linear, quadratic, and exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9

f(x) = 2x + 1 and g has table g(0) = 3, g(1) = 4. Which has the greater y-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q1/5

$f(x) = x^2$ and $g(x) = 2^x$. Which is greater at $x = 10$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q2/5

True or false?

An increasing exponential eventually exceeds any linear function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q3/5

$f(x) = -(x - 1)^2 + 5$ and g with maximum 4 in a graph. The greater maximum is

A. f: 5

B. cannot tell

C. equal

D. g: 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q4/5

A table rises by 5 each step. What kind of function is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 2

Recognize a pattern in a table • Grade 8 • Functions • 8.F.A.3

Is the table x: 0, 1, 2; y: 10, 15, 20 linear? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q1/5

Is the table x: 1, 3, 9, 27; y: 0, 1, 2, 3 linear? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q2/5

True or false?

A table with constant ratios of outputs is exponential.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q3/5

Which table is exponential?

A. 1, 4, 9, 16

B. 1, 2, 4, 8

C. 2, 2, 2, 2

D. 1, 3, 5, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q4/5

What is the slope of a linear table with outputs 10, 15, 20 at x = 0, 1, 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 3

Domains, asymptotes, and periods of Algebra II families • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the domain of $\sqrt{(x - 1)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q1/5

Give the asymptotes of $(2x + 1)/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q2/5

True or false?

The period of $12 + 10\sin(\pi t/15)$ is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q3/5

The domain of $\log_2 x$ is

A. all reals

B. $x \geq 0$

C. $x > 0$

D. $x \neq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q4/5

A wheel of radius 8 m with center 10 m high has what maximum height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • HSF-IF.C.9

FORWARD 1

Compare rates of change and long-run behavior across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = 3(1.1)^x$ and $g(x) = 10 + 5x$. Estimate the first whole number x at which f exceeds g. A calculator may be used. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q1/5

$f(x) = \log_2 x$, and g has $g(1) = 0$, $g(3) = 1$, $g(9) = 2$. Compare the average rates of change of f and g on $1 \le x \le 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q2/5

True or false?

A function with a larger y-intercept is always larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q3/5

Which grows fastest for large x?

A. $\sqrt{(x - 1)}$

B. $3(1.1)^x$

C. $5x + 10$

D. $\log_2 x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q4/5

Explain why $g(x) = \log_3 x$ stays below $f(x) = \log_2 x$ for $x > 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSF-IF.C.9

FORWARD 2

Build a difference function to compare • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

Write $d(x) = f(x) - g(x)$ for $f(x) = 3(1.1)^x$ and $g(x) = 5x + 10$.

$d(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q1/5

Explain what the sign of d(x) tells you.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q2/5

True or false?

A zero of $d(x) = f(x) - g(x)$ is an x where $f(x) = g(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q3/5

For $h_1(t) = 12 + 10\sin(\pi t/15)$ and $h_2(t) = 10 + 8\sin(\pi t/20)$, $h_1(0) - h_2(0)$ is

A. 2

B. 22

C. -2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q4/5

Two Ferris wheel heights are $h_1(t) = 12 + 10\sin(\pi t/15)$ and $h_2(t) = 10 + 8\sin(\pi t/20)$. Write $h_1(t) - h_2(t)$ as a function of t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q5/5

Build models by combining Algebra II function types

MAIN	Algebra II main question	**HSF-BF.A.1.b**
BACKWARD 1	Combine linear and quadratic functions	**HSF-BF.A.1.b**
BACKWARD 2	Multiply and subtract polynomials	**HSA-APR.A.1**
BACKWARD 3	Evaluate exponential and trigonometric expressions	**HSF-IF.A.2**
FORWARD 1	Interpret the parts of a combined model	**HSF-BF.A.1.b**
FORWARD 2	Compose functions in context	**HSF-BF.A.1.c**

SET 54 • HSF-BF.A.1.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

Revenue is $R(x) = x(50 - 0.5x)$ and cost is $C(x) = 200 + 10x$ dollars for x items. Write the profit $P(x)$ in standard form and find $P(30)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q1/5

The cost of x posters is C(x) = 1200 + 3x dollars. Write the average cost per poster A(x) and find A(100).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q2/5

A town has $P(t) = 5000(1.02)^t$ people and $F(t) = 6000 + 100t$ units of food after t years. Write the food per person and find it at t = 10 to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q3/5

A spring's amplitude is 10 cm and shrinks by 10% each second, while it oscillates once per second. Write y(t) as an exponential times a cosine and find y(1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q4/5

Coffee starts 70°C above a 20°C room, and the difference shrinks 5% each minute. Write its temperature T(t) as a sum of functions and find T(10) to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 1

Combine linear and quadratic functions • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

$R(x) = 8x$ and $C(x) = 3x + 50$. Write $P = R - C$.

$P(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q1/5

$f(x) = x + 2$ and $g(x) = x - 2$. Write $f \cdot g$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q2/5

True or false?

Profit is revenue minus cost.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q3/5

Average cost is

A. total cost times items

B. total cost divided by the number of items

C. cost minus revenue

D. revenue divided by cost

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q4/5

$f(x) = 2x$ and $g(x) = x^2 + 1$. Find $(f + g)(3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 2

Multiply and subtract polynomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $x(50 - 0.5x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q1/5

Simplify $(50x - 0.5x^2) - (200 + 10x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q2/5

True or false?

$-(200 + 10x) = -200 + 10x.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q3/5

$(3x + 1) - (x - 4)$ equals

A. $2x + 5$

B. $4x - 3$

C. $2x - 3$

D. $2x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q4/5

Expand $2(x^2 - 3x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 3

Evaluate exponential and trigonometric expressions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Evaluate $70(0.95)^{10}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q1/5

Evaluate $5000(1.02)^{10}$ to the nearest whole number. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q2/5

True or false?

$\cos(2\pi) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q3/5

$10(0.9)^2$ equals

A. 9

B. 18

C. 81

D. 8.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q4/5

Evaluate $\cos(2\pi \cdot 0.5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q5/5

SET 54 • HSF-BF.A.1.b

FORWARD 1

Interpret the parts of a combined model • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

In $T(t) = 20 + 70(0.95)^t$, what do 20 and $70(0.95)^t$ represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q1/5

In $y(t) = 10(0.9)^t \cos(2\pi t)$, which factor controls the size of the swings and which controls their timing?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q2/5

True or false?

In $A(x) = (1200 + 3x)/x = 1200/x + 3$, the term $1200/x$ is the fixed cost spread over x posters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q3/5

As t grows, $T(t) = 20 + 70(0.95)^t$ approaches

A. 20

B. 0

C. 90

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q4/5

Explain why food per person F(t)/P(t) eventually decreases.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSF-BF.A.1.b

FORWARD 2

Compose functions in context • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

A balloon's radius is $r(t) = 2t$ cm after t seconds and its volume is $V(r) = (4/3)\pi r^3$. Write $V(r(t))$.

Answer: _____

A store takes 20% off with $d(p) = 0.8p$ and then adds a \$5 fee with $f(p) = p + 5$. Write $f(d(p))$ and find it for $p = 50$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q2/5

True or false?

f(g(x)) and g(f(x)) are always equal.

A. True

B. False

If $f(x) = x^2$ and $g(x) = x + 1$, then $f(g(2)) =$

A. 4

B. 9

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q4/5

Explain how composition differs from multiplying two functions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q5/5

Identify vertical shifts, vertical stretches, and horizontal shifts of Algebra II functions

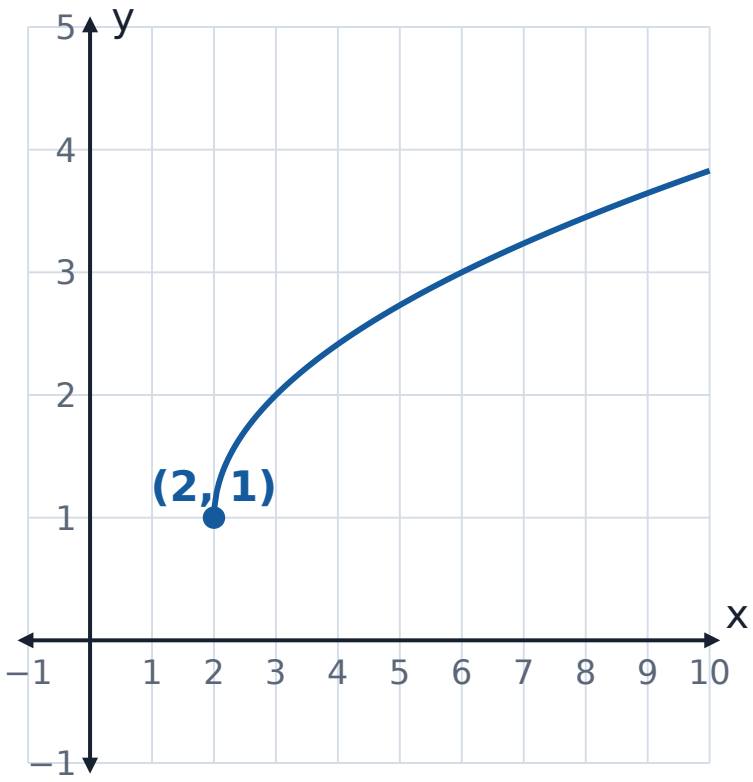
MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Shifts and stretches of linear and quadratic graphs	**HSF-BF.B.3**
BACKWARD 2	Translate points in the plane	**8.G.A.3**
BACKWARD 3	Key points of parent functions	**HSF-IF.C.7.e**
FORWARD 1	Write transformed functions from descriptions and order of transfo..	**HSF-BF.B.3**
FORWARD 2	Read sinusoid parameters as transformations	**HSF-TF.B.5**

SET 55 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

$f(x) = \sqrt{x}$. The graph of $g(x) = f(x - h) + k$ is shown. Find h and k.



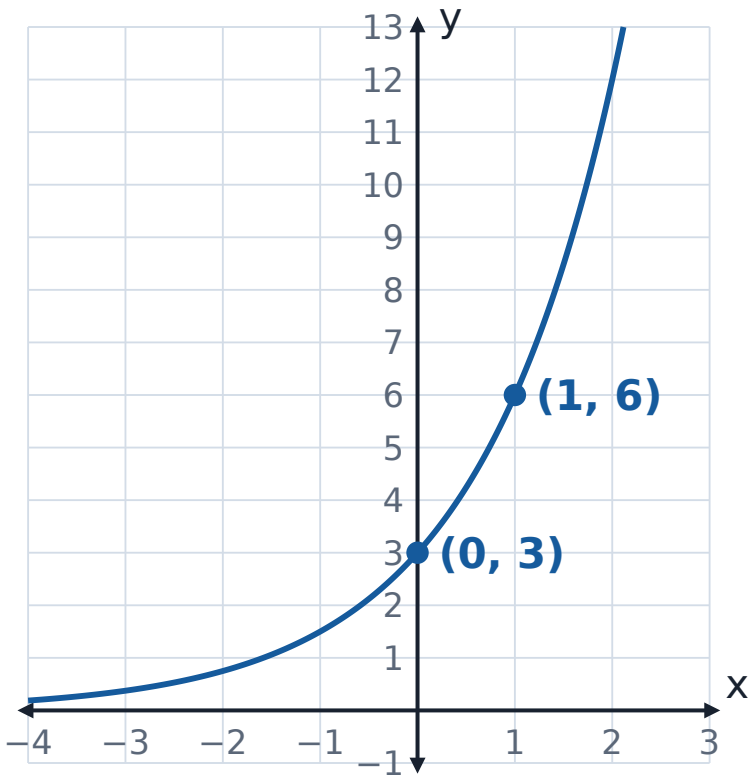
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q1/5

$f(x) = 2^x$. The graph of $g(x) = a \cdot f(x)$ is shown. Find a.



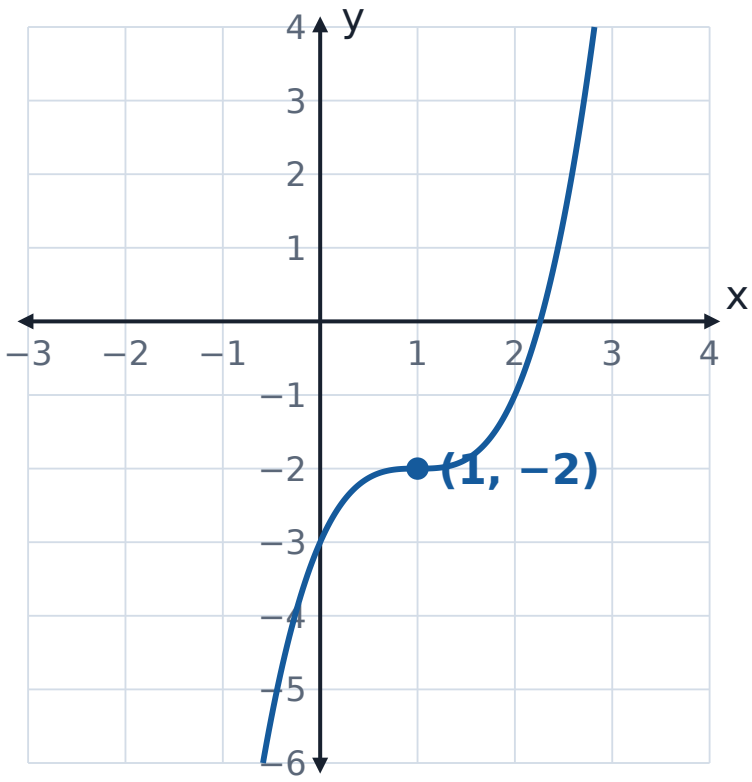
a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q2/5

$f(x) = x^3$. The graph of $g(x) = f(x - h) + k$ is shown. Find h and k.



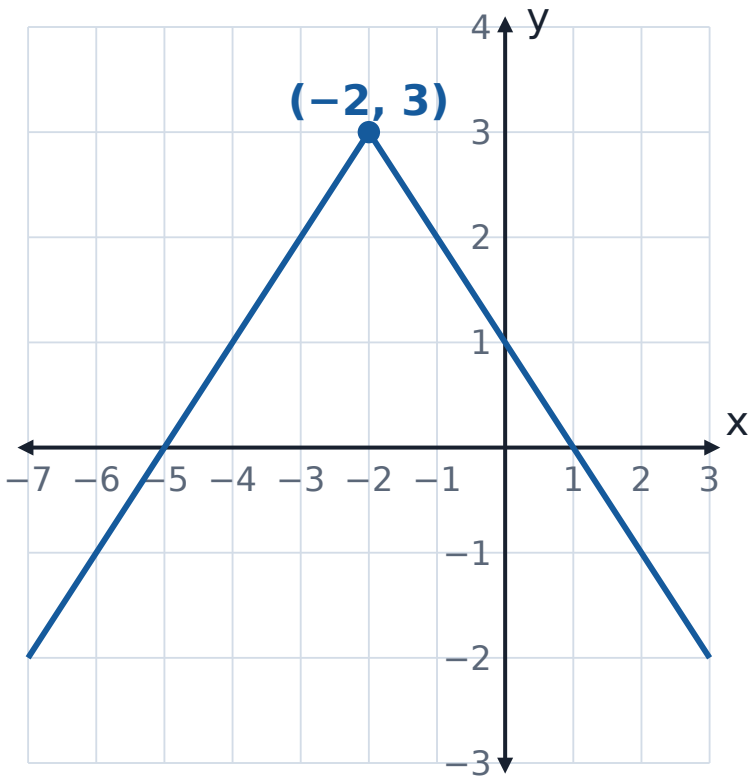
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q3/5

$f(x) = |x|$. The graph of $g(x) = a \cdot f(x - h) + k$ is shown. Find a, h, and k.



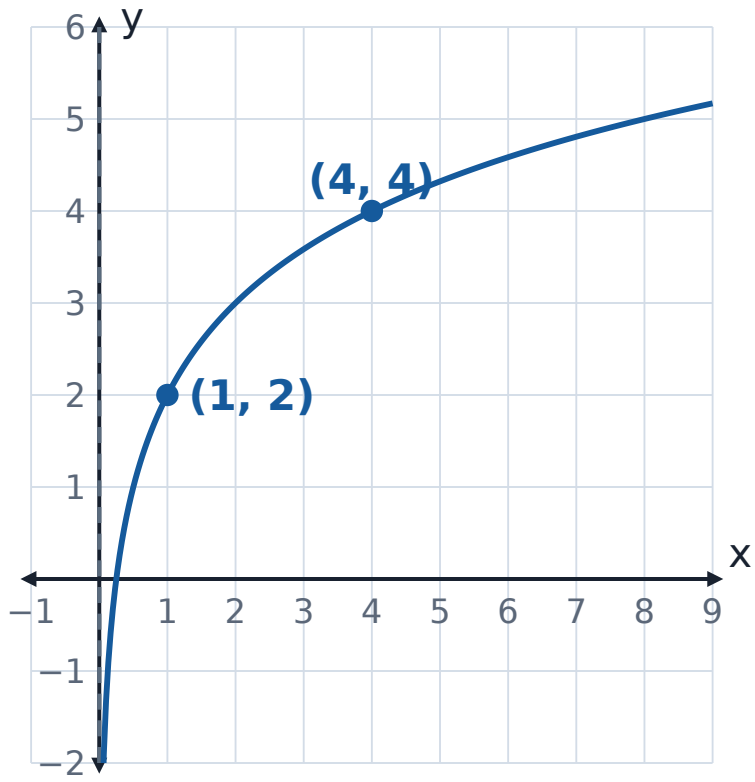
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q4/5

$f(x) = \log_2 x$. The graph of $g(x) = f(x) + k$ is shown. Find k.



$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 1

Shifts and stretches of linear and quadratic graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is $y = (x - 3)^2 + 1$ obtained from $y = x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q1/5

How is $y = 2x^2$ obtained from $y = x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q2/5

True or false?

$y = f(x + 4)$ is the graph of f shifted right 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q3/5

The vertex of $y = (x + 2)^2 - 5$ is

A. (2, 5)

B. (2, −5)

C. (−2, −5)

D. (−2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q4/5

Write $y = x^2$ shifted left 1 and down 4.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 2

Translate points in the plane • Grade 8 • Geometry • 8.G.A.3

Translate (0, 0) right 2 and up 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q1/5

Translate (1, 1) left 2 and up 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q2/5

True or false?

Translating (0, 0) down 2 gives (−2, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q3/5

$(x, y) \to (x - 2, y + 3)$ moves $(0, 0)$ to

A. $(-2, 3)$

B. $(-2, -3)$

C. $(2, 3)$

D. $(3, -2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q4/5

Which translation takes (0, 0) to (1, −2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 3

Key points of parent functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Give the point of $y = \sqrt{x}$ at $x = 0$ and at $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q1/5

Give the y-intercept of $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q2/5

True or false?

The graph of $y = \log_2 x$ passes through (1, 0).

A. True

B. False

The vertex of $y = |x|$ is

A. (1, 0)

B. (1, 1)

C. (0, 1)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q4/5

Give the center (point of symmetry) of $y = x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q5/5

SET 55 • HSF-BF.B.3

FORWARD 1

Write transformed functions from descriptions and order of transformations • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Write g, the graph of $f(x) = 2^x$ stretched vertically by 3, then shifted down 4.

$g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q1/5

Write g, the graph of $f(x) = \sqrt{x}$ shifted down 4, then stretched vertically by 3.

$g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q2/5

True or false?

A vertical stretch then a shift can give a different graph than the same shift then the stretch.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q3/5

$g(x) = 2f(x) + 1$ moves the point (3, 4) of f to

A. (6, 5)

B. (4, 9)

C. (3, 9)

D. (3, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q4/5

$g(x) = f(x + 2) - 3$ and f has asymptote $x = 0$. Where is g's vertical asymptote?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSF-BF.B.3

FORWARD 2

Read sinusoid parameters as transformations • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Describe $y = 3\sin x + 2$ as transformations of $y = \sin x$ and give its maximum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q1/5

Write the function for y = cos x stretched vertically by 4 and shifted down 1.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q2/5

True or false?

In $y = A \sin x + D$, A is the midline.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q3/5

$y = -2\cos x + 5$ has amplitude and midline

A. 5 and $y = 2$

B. -2 and $y = 5$

C. 2 and $y = -5$

D. 2 and $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q4/5

A model is $y = A \sin x + D$ with maximum 9 and minimum 1. Find A ($A > 0$) and D.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q5/5

Horizontal stretches, compressions, and reflections f(kx)

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Reflections and dilations of points	**8.G.A.3**
BACKWARD 2	Solve kx = c for x	**6.EE.B.7**
BACKWARD 3	Period of y = sin(Bx)	**HSF-TF.B.5**
FORWARD 1	Combine horizontal stretches with shifts	**HSF-BF.B.3**
FORWARD 2	Graph exponential and trigonometric functions with horizontal cha…	**HSF-IF.C.7.e**

SET 56 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

$f(x) = \sqrt{x}$ passes through (4, 2). Describe the graph of $g(x) = \sqrt{(4x)}$ compared with f, and give the point of g with y = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q1/5

$f(x) = \sin x$ has period 2π. Describe $g(x) = \sin(3x)$ compared with f and give its period.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q2/5

Describe the graph of $g(x) = 2^{-x}$ compared with $f(x) = 2^{x}$, and give the point of g that comes from (1, 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q3/5

f passes through $(-2, 1)$, $(0, 3)$, and $(4, -1)$. List the corresponding points of $g(x) = f(2x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q4/5

f passes through (2, 5). Which point must be on the graph of g(x) = f(−x/2)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 1

Reflections and dilations of points • Grade 8 • Geometry • 8.G.A.3

Reflect (1, 2) over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q1/5

Multiply the x-coordinate of $(4, -1)$ by $1/2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q2/5

True or false?

Reflecting over the y-axis changes the sign of the x-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q3/5

(x, y) → (x/3, y) applied to (6, 2) gives

A. (18, 2)

B. (6, 2/3)

C. (2, 2)

D. (3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q4/5

Apply $(x, y) \to (-2x, y)$ to $(2, 5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 2

Solve kx = c for x • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve 2x = 4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q1/5

Solve $-x/2 = 2$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q2/5

True or false?

If $4x = 4$, then $x = 16$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q3/5

$3x = 2\pi$ gives $x =$

A. $2\pi/3$

B. $2\pi - 3$

C. $\pi/3$

D. 6π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q4/5

Solve $-x = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 3

Period of y = sin(Bx) • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Give the period of $y = \sin(2x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q1/5

Give the period of $y = \cos(x/2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q2/5

True or false?

For $y = \sin(Bx)$ with $B > 0$, a larger B gives a longer period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q3/5

$y = \sin(4x)$ has period

A. 8π

B. 4π

C. $\pi/4$

D. $\pi/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q4/5

Which B gives period 6?

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q5/5

SET 56 • HSF-BF.B.3

FORWARD 1

Combine horizontal stretches with shifts • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Rewrite $g(x) = f(2x - 6)$ in the form $f(k(x - h))$ and describe the transformation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q1/5

f passes through (4, 1). Which point is on $g(x) = f(2x - 6)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q2/5

True or false?

$f(2x - 6)$ is $f(2x)$ shifted right 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q3/5

$y = \sqrt{(9x)}$ equals

A. $\sqrt{x} + 9$

B. $9\sqrt{x}$

C. $3\sqrt{x}$

D. $\sqrt{x}/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q4/5

Write g, the graph of $f(x) = 2^x$ reflected over the y-axis and then shifted right 1.

$g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSF-BF.B.3

FORWARD 2

Graph exponential and trigonometric functions with horizontal changes • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Give the y-intercept, asymptote, and two points of $y = 2^{-x} + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q1/5

Give the maxima of $y = \cos(2x)$ on $0 \leq x \leq 2\pi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q2/5

True or false?

$y = 3^{2x} = 9^{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q3/5

$y = \sin(x/2)$ completes one cycle on

A. $0 \leq x \leq \pi$

B. $0 \leq x \leq 2\pi$

C. $0 \leq x \leq \pi/2$

D. $0 \leq x \leq 4\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q4/5

Explain why $y = 2^{2x}$ is a horizontal compression of $y = 2^{x}$ and also an exponential with base 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q5/5

Recognize even and odd functions

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Signs of powers of negatives	**7.NS.A.2**
BACKWARD 2	Reflect points over the axes and the origin	**6.NS.C.6.b**
BACKWARD 3	Substitute -x into an expression	**HSF-IF.A.2**
FORWARD 1	Reason about sums and products of even and odd functions	**HSF-BF.B.3**
FORWARD 2	Explain trigonometric symmetry with the unit circle	**HSF-TF.A.4**

SET 57 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Is $f(x) = x^4 - 3x^2$ even, odd, or neither? Show $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q1/5

Is $f(x) = x^3 - x$ even, odd, or neither? Show $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q2/5

Is $f(x) = x^2 + x$ even, odd, or neither? Justify.

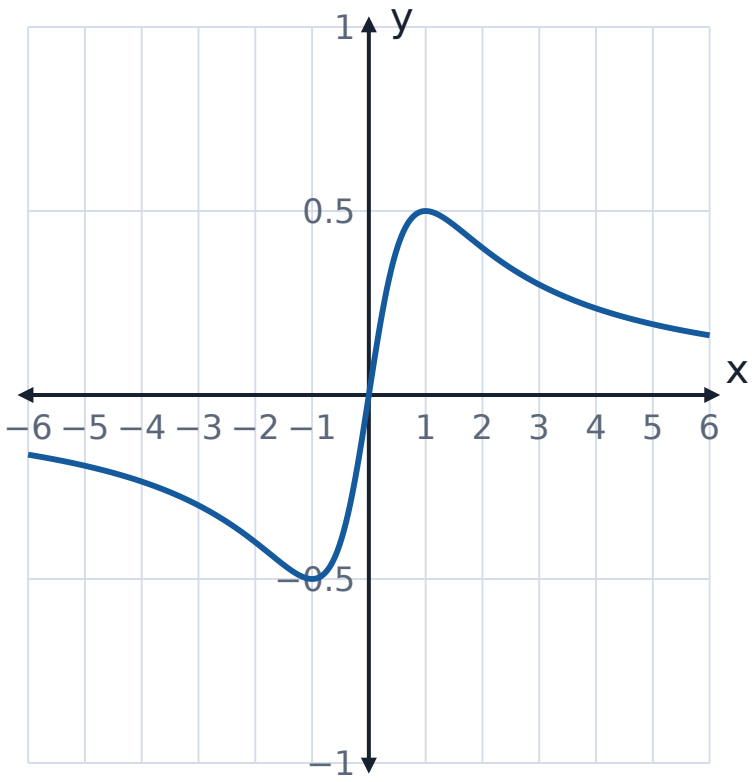
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q3/5

The graph of $f(x) = x/(x^2 + 1)$ is shown. Is f even, odd, or neither? Explain from the graph.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q4/5

Is $f(x) = \cos x + x^2$ even, odd, or neither? Show $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 1

Signs of powers of negatives • Grade 7 • The Number System • 7.NS.A.2

Simplify $(-x)^4$ and $(-x)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q1/5

Simplify $(-x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q2/5

True or false?

$(-x)^5 = x^5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q3/5

$(-2)^2 - (-2)$ equals

A. -6

B. 2

C. 6

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q4/5

Simplify $-(-x)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 2

Reflect points over the axes and the origin • Grade 6 • The Number System • 6.NS.C.6.b

Reflect (2, 3) over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q1/5

Rotate (2, 3) 180° about the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q2/5

True or false?

A graph symmetric about the y-axis contains $(-a, b)$ whenever it contains (a, b).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q3/5

Symmetry about the origin pairs (a, b) with

A. (−a, b)

B. (−a, −b)

C. (a, −b)

D. (b, a)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q4/5

A graph is symmetric about the origin and contains (1, 0.5). Name another point.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 3

Substitute −x into an expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

For $f(x) = x^2 + x$, find $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q1/5

For $f(x) = x^3 - x$, find $f(-2)$ and $f(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q2/5

True or false?

For $f(x) = 3x$, $f(-x) = 3x$.

A. True

B. False

For $f(x) = x^4 - 3x^2$, $f(-1)$ is

A. 2

B. 4

C. -2

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q4/5

For $f(x) = 1/x$, find $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • HSF-BF.B.3

FORWARD 1

Reason about sums and products of even and odd functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Prove that the sum of two odd functions is odd.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q1/5

Prove that the product of two odd functions is even.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q2/5

True or false?

The only function that is both even and odd is $f(x) = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q3/5

$f(x) = |x| + x^3$ is

A. even

B. both

C. neither

D. odd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q4/5

Is f(x) = x sin x even, odd, or neither? Explain using the fact that a product of two odd functions is even.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSF-BF.B.3

FORWARD 2

Explain trigonometric symmetry with the unit circle • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Use the unit circle to explain why $\cos(-x) = \cos x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q1/5

Is tan even, odd, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q2/5

True or false?

sin is an even function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q3/5

Which is an even function?

A. cos x

B. tan x

C. sin x + cos x

D. sin x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q4/5

Use symmetry to find $\sin(-\pi/6)$ from $\sin(\pi/6) = 1/2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q5/5

Write inverses of radical, rational, cubic, exponential, and logarithmic functions

MAIN	Algebra II main question	**HSF-BF.B.4.a**
BACKWARD 1	Inverses of linear functions	**HSF-BF.B.4.a**
BACKWARD 2	Rearrange formulas with roots, powers, and fractions	**HSA-CED.A.4**
BACKWARD 3	Domain and range	**HSF-IF.B.5**
FORWARD 1	Inverses that are their own inverse, and checking by points	**HSF-BF.B.4.a**
FORWARD 2	Verify inverses by composition	**HSF-BF.B.4.b**

SET 58 • HSF-BF.B.4.a

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

Find the inverse of $f(x) = (x - 1)^3 + 2$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q1/5

Find the inverse of $f(x) = 2x/(x - 3)$, $x \neq 3$, and state its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q2/5

Find the inverse of $f(x) = \sqrt{(x + 4)}$, $x \geq -4$, and state its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q3/5

Find the inverse of $f(x) = 5 \cdot 2^x$ and state its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q4/5

Find the inverse of $f(x) = \log_3(x - 1)$ and state its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 1

Inverses of linear functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

Find the inverse of $f(x) = 3x - 6$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

Solve $2x + 5 = y$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

True or false?

The inverse undoes the function: $f^{-1}(f(x)) = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

The inverse of $f(x) = x/4 + 1$ is

A. $4(x - 1)$

B. $(x - 1)/4$

C. $4x - 1$

D. $x/4 - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

If $f(2) = 7$, what is $f^{-1}(7)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 2

Rearrange formulas with roots, powers, and fractions • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve $y = \sqrt{(x + 4)}$ for x ($y \geq 0$).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q1/5

Solve $y = 2x/(x - 3)$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q2/5

True or false?

Solving $y = (x - 1)^3 + 2$ for x gives $x = \sqrt[3]{(y - 2)} + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q3/5

Solving $y = x^3$ for x gives

A. $y/3$

B. $\sqrt[3]{y}$

C. $\sqrt{y}$

D. y^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q4/5

Solve $y = x/(x + 1)$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 3

Domain and range • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

Give the range of $f(x) = \sqrt{(x + 4)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q1/5

Give the range of $f(x) = 5 \cdot 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q2/5

True or false?

The range of $f(x) = 2x/(x - 3)$ excludes $y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q3/5

The domain of $\log_3(x - 1)$ is

A. $x > 1$

B. all reals

C. $x > 0$

D. $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q4/5

Give the range of $f(x) = \log_3(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q5/5

SET 58 • HSF-BF.B.4.a

FORWARD 1

Inverses that are their own inverse, and checking by points • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

Find the inverse of $f(x) = (x + 1)/(x - 1)$. What do you notice?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q1/5

$f(x) = 5 \cdot 2^x$ passes through (1, 10), and $f^{-1}(x) = \log_2(x/5)$. Check the inverse with this point.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q2/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q3/5

Which is its own inverse?

A. $f(x) = x^2$

B. $f(x) = x + 1$

C. $f(x) = 2x$

D. $f(x) = 1/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q4/5

Use the horizontal line test on the graph of $f(x) = x^3 - x$ to decide whether f has an inverse function. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSF-BF.B.4.a

FORWARD 2

Verify inverses by composition • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

Verify that $f(x) = (x - 1)^3 + 2$ and $g(x) = \sqrt[3]{(x - 2)} + 1$ are inverses by computing $f(g(x))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q1/5

$f(x) = (x - 1)^3 + 2$ and $g(x) = \sqrt[3]{(x - 2)} + 1$. Compute $g(f(x))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q2/5

True or false?

To verify inverses, both $f(g(x)) = x$ and $g(f(x)) = x$ must hold on the domains.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q3/5

For $f(x) = \sqrt{(x + 4)}$ and $g(x) = x^2 - 4$ $(x \geq 0)$, $g(f(x)) =$

A. x for all x

B. $|x|$

C. x for $x \geq -4$

D. $x^2 + 4$

Explain why $g(x) = x^2 - 4$ without a domain restriction is not the inverse of $f(x) = \sqrt{(x + 4)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q5/5

Write exact logarithmic solutions of exponential equations

MAIN	Algebra II main question	**HSF-LE.A.4**
BACKWARD 1	Isolate the exponential expression	**HSA-REI.B.3**
BACKWARD 2	Powers of 2, 3, and 10	**8.EE.A.1**
BACKWARD 3	Logarithms as the inverse of exponentials	**HSF-BF.B.4.a**
FORWARD 1	Solve equations with different bases or variables on both sides	**HSF-LE.A.4**
FORWARD 2	Use the inverse relationship to solve logarithmic equations	**HSF-BF.B.5**

SET 59 • HSF-LE.A.4

MAIN

Algebra II main question • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $3 \cdot 2^{5t} = 48$ exactly.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q1/5

Solve $2 \cdot 10^{3t} = 9$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q2/5

Solve $5e^{2t} = 30$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q3/5

Solve $100(1.05)^t = 300$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q4/5

Solve $4 \cdot 3^{t/2} = 20$ exactly.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 1

Isolate the exponential expression • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 3y = 48 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q1/5

Solve $5y = 30$ for y.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q2/5

True or false?

From $2 \cdot 10^{3t} = 9$, we get $10^{3t} = 4.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q3/5

From $4 \cdot 3^{t/2} = 20$, $3^{t/2}$ equals

A. 80

B. 5

C. 1/5

D. 16

Solve 100y = 300 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 2

Powers of 2, 3, and 10 • Grade 8 • Expressions & Equations • 8.EE.A.1

Write 16 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q1/5

Write 1000 as a power of 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q2/5

True or false?

$3^4 = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q3/5

$2^{5t} = 2^4$ gives

A. t = 4

B. 5t = 4

C. 5t = 16

D. t = 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q4/5

Write 1/8 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 3

Logarithms as the inverse of exponentials • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

Rewrite $10^x = 4.5$ as a logarithm.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q1/5

Rewrite $e^{u} = 6$ as a logarithm.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q2/5

True or false?

$\log_2 8 = 4$ because $2^4 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q3/5

$3^y = 5$ means y =

A. $\log_3 5$

B. 5/3

C. $\log_5 3$

D. 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q4/5

Evaluate $\log_2 32$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q5/5

SET 59 • HSF-LE.A.4

FORWARD 1

Solve equations with different bases or variables on both sides • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $2^{x} = 3^{x-1}$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q1/5

Solve $5^{2x} = 7$ exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q2/5

True or false?

$\log(ab) = \log a + \log b$ for positive a and b.

A. True

B. False

The exact solution of $e^{x+1} = 10$ is

A. 10/e

B. ln 10/1

C. ln 9

D. ln 10 − 1

Solve $3 \cdot 4^x = 2 \cdot 5^x$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSF-LE.A.4

FORWARD 2

Use the inverse relationship to solve logarithmic equations • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

Solve $\log_2(x + 3) = 5$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q1/5

Solve $\log x + \log(x - 3) = 1$ and check for extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q2/5

True or false?

$\log_b(b^x) = x$ for every real x.

A. True

B. False

$\ln(e^7)$ equals

A. $\ln 7$

B. 1

C. 7

D. e^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q4/5

Solve $\ln(2x) = 3$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q5/5

Evaluate logarithmic solutions of exponential models with technology

MAIN	Algebra II main question	**HSF-LE.A.4**
BACKWARD 1	Write exact logarithmic solutions	**HSF-LE.A.4**
BACKWARD 2	Evaluate logarithms with a calculator	**HSF-IF.A.2**
BACKWARD 3	Choose an appropriate level of accuracy	**HSN-Q.A.3**
FORWARD 1	Analyze how parameters change the solution	**HSF-LE.A.4**
FORWARD 2	Rewrite models to show doubling time or half-life	**HSF-IF.C.8.b**

SET 60 • HSF-LE.A.4

MAIN

Algebra II main question • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$2,500 earns 3.5% interest compounded yearly. Solve $2500(1.035)^t = 4000$ and interpret the answer. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q1/5

Carbon-14 has a half-life of 5730 years. Solve $80(0.5)^{t/5730} = 30$ to the nearest year and interpret it. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q2/5

Coffee cools by $T(t) = 20 + 70(0.95)^t$ °C after t minutes. When is it 50°C? Give minutes to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q3/5

A city of 1.2 million grows by $P(t) = 1.2e^{0.015t}$ million people. When does it reach 1.5 million? Give years to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q4/5

A dose leaves $200(0.8)^t$ mg of a drug after t hours. After which whole hour is the amount first below 50 mg? A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 1

Write exact logarithmic solutions • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $1.035^{t} = 1.6$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q1/5

Solve $e^{0.015t} = 1.25$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q2/5

True or false?

$0.8^t = 0.25$ gives $t = \log 0.25/\log 0.8$.

A. True

B. False

$(0.5)^{t/5730} = 0.375$ gives t =

A. $0.375 \cdot 5730$

B. 5730/log 0.375

C. log 0.375/5730

D. 5730 log 0.375/log 0.5

Solve $0.95^{t} = 3/7$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 2

Evaluate logarithms with a calculator • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Evaluate log 1.6 to four decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q1/5

Evaluate ln 1.25 to four decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q2/5

True or false?

log 0.25 is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q3/5

log 1.6/log 1.035 is about

A. 0.20

B. 46.0

C. 1.55

D. 13.66

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q4/5

Evaluate log 0.25/log 0.8 to two decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 3

Choose an appropriate level of accuracy • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

A model gives 13.66 years for yearly interest. What whole-year answer fits the context?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q1/5

Round 8108.16 years to the nearest hundred years.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q2/5

True or false?

A time found from a model should be reported with more decimal places than the data justify.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q3/5

A drug amount first drops below 50 mg at t = 6.21 h. If it is checked every hour, it is first below 50 mg at

A. 6.21 h

B. 6 h

C. 7 h

D. 5 h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q4/5

Why might 14.88 years be reported as "about 15 years"?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • HSF-LE.A.4

FORWARD 1

Analyze how parameters change the solution • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

If the interest rate changes from 3.5% to 5%, how long does it take $2,500 to reach $4,000? Give years to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q1/5

Find the doubling time at 6% yearly interest and compare it with the Rule of 72 estimate 72/6. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q2/5

True or false?

Doubling the initial amount in $a(b)^t = d$ halves the time t needed.

A. True

B. False

In $200(0.8)^t = 50$, replacing 200 by 400 changes t by

A. doubling it

B. halving it

C. adding $\log 2/\log(1/0.8) \approx 3.11$ hours

D. nothing

At 3.5% yearly interest, explain why $2,500 takes as long to grow to $4,000 as $5,000 takes to grow to $8,000.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSF-LE.A.4

FORWARD 2

Rewrite models to show doubling time or half-life • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Rewrite $2500(1.035)^t$ in the form $2500 \cdot 2^{t/D}$ and give D to the nearest hundredth. A calculator may be used.

$D \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q1/5

Rewrite $200(0.8)^t$ in the form $200(0.5)^{t/H}$ and give H to the nearest hundredth. A calculator may be used.

$H \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q2/5

True or false?

$80(0.5)^{t/5730}$ shows the half-life 5730 directly.

A. True

B. False

$1.2e^{0.015t}$ has doubling time about

A. 46.2 years

B. 66.7 years

C. 15 years

D. 0.015 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q4/5

Explain why the half-life form makes it easy to answer "how much remains after 3 half-lives".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSS-ID.A.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate population percentages with a normal distribution

MAIN	Algebra II main question	**HSS-ID.A.4**
BACKWARD 1	Interpret mean and standard deviation	**HSS-ID.A.2**
BACKWARD 2	Describe the shape of a distribution	**HSS-ID.A.1**
BACKWARD 3	Percents of a whole	**6.RP.A.3.c**
FORWARD 1	Find values from percentiles of a normal model	**HSS-ID.A.4**
FORWARD 2	Use a normal model for a margin of error	**HSS-IC.B.4**

SET 61 • HSS-ID.A.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Adult women's heights are approximately normal with mean 64 in and standard deviation 2.5 in. Estimate the percent between 59 and 69 inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q1/5

Adult women's heights are approximately normal with mean 64 in and standard deviation 2.5 in. Estimate the percent of women taller than 66.5 inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q2/5

Test scores are approximately normal with mean 500 and standard deviation 100. Estimate the percent of scores between 400 and 700. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q3/5

Battery life is approximately normal with mean 40 h and standard deviation 4 h. Use the table to estimate the percent of batteries lasting less than 46 hours.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q4/5

Cereal box weights are approximately normal with mean 16.0 oz and standard deviation 0.2 oz. Use the table to estimate the percent between 15.8 and 16.3 oz.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 1

Interpret mean and standard deviation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

A data set has mean 64 and standard deviation 2.5. What value is one standard deviation above the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q1/5

Which value is two standard deviations below a mean of 500 with standard deviation 100?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q2/5

True or false?

A larger standard deviation means the data are more spread out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q3/5

46 is how many standard deviations above a mean of 40 when the standard deviation is 4?

A. 0.67

B. 4

C. 6

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q4/5

Compute $(15.8 - 16.0)/0.2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 2

Describe the shape of a distribution • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Describe a histogram that is highest in the middle and falls off evenly on both sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q1/5

Where are the mean and median of a symmetric distribution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q2/5

True or false?

A normal curve is symmetric about its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q3/5

A bell-shaped histogram is called

A. skewed right

B. uniform

C. mound-shaped and symmetric

D. bimodal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q4/5

In a symmetric distribution, what percent of the data are below the median?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 3

Percents of a whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Find 95% of 2,000.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q1/5

What is 34% + 13.5%?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q2/5

True or false?

Half of 68% is 36%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q3/5

100% − 95% split equally into two tails gives each tail

A. 5%

B. 47.5%

C. 0.5%

D. 2.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q4/5

Write 0.7745 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q5/5

SET 61 • HSS-ID.A.4

FORWARD 1

Find values from percentiles of a normal model • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Above what height are the tallest 2.5% of women (mean 64 in, SD 2.5 in)? Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

Use the table: what battery life is the 84th percentile (mean 40 h, SD 4 h)?

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

True or false?

In a normal model, the median equals the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

The 16th percentile of scores with mean 500 and SD 100 is

A. 484

B. 600

C. 340

D. 400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

A cereal box below 15.7 oz is rejected. Use the table to estimate the percent rejected (mean 16.0, SD 0.2).

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSS-ID.A.4

FORWARD 2

Use a normal model for a margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

Simulated sample proportions are approximately normal with mean 0.58 and SD 0.049. Within what interval do about 95% of them fall?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q1/5

Simulated sample proportions have standard deviation 0.049. What margin of error does this give for the population proportion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q2/5

True or false?

The margin of error is about two standard deviations of the simulated statistic.

A. True

B. False

If the SD of simulated means is 0.2 hours, the margin of error is about

A. 0.1 hours

B. 0.4 hours

C. 2 hours

D. 0.2 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q4/5

Explain why the normal model is useful for simulated sample proportions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSS-ID.A.4 • STATISTICS, INFERENCE, AND DECISIONS

Recognize data sets for which a normal model is not appropriate

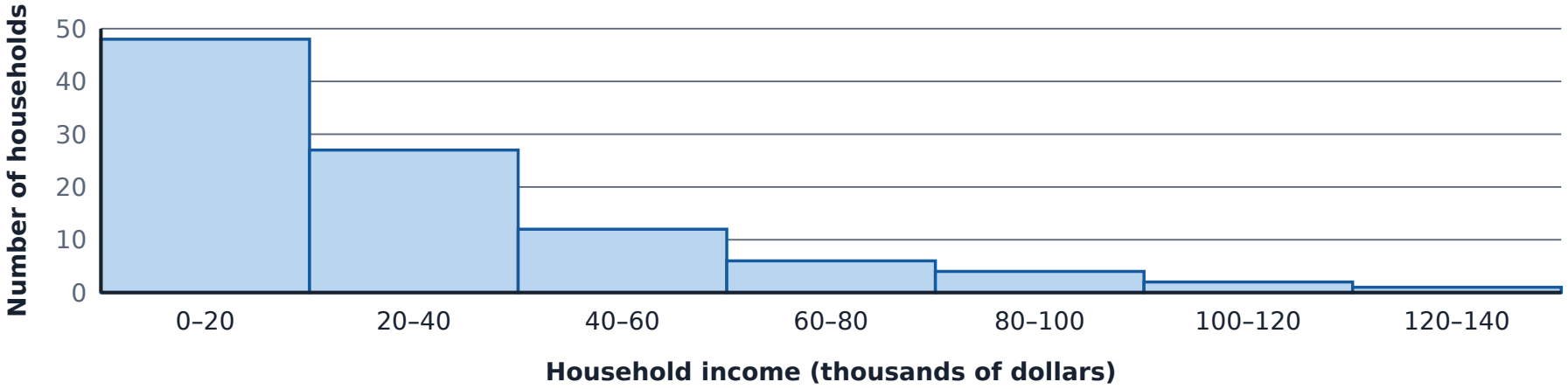
MAIN	Algebra II main question	**HSS-ID.A.4**
BACKWARD 1	Describe skewed, symmetric, bimodal, and uniform shapes	**HSS-ID.A.3**
BACKWARD 2	Measures of center and spread for a data set	**6.SP.A.2**
BACKWARD 3	Groups combined in one data set	**HSS-ID.A.1**
FORWARD 1	Check a normal model with the 68-95-99.7 rule	**HSS-ID.A.4**
FORWARD 2	Decide whether a probability model fits observed data	**HSS-IC.A.2**

SET 62 • HSS-ID.A.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

The histogram shows household incomes in a town. Is a normal model appropriate? Justify.



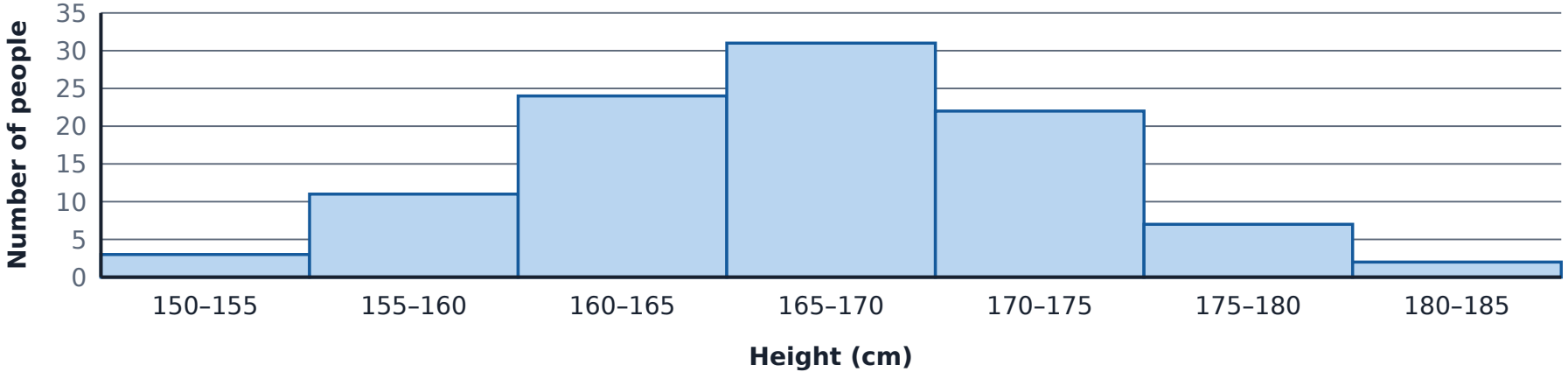
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q1/5

The histogram shows heights of adults in a club. Is a normal model appropriate? Justify.



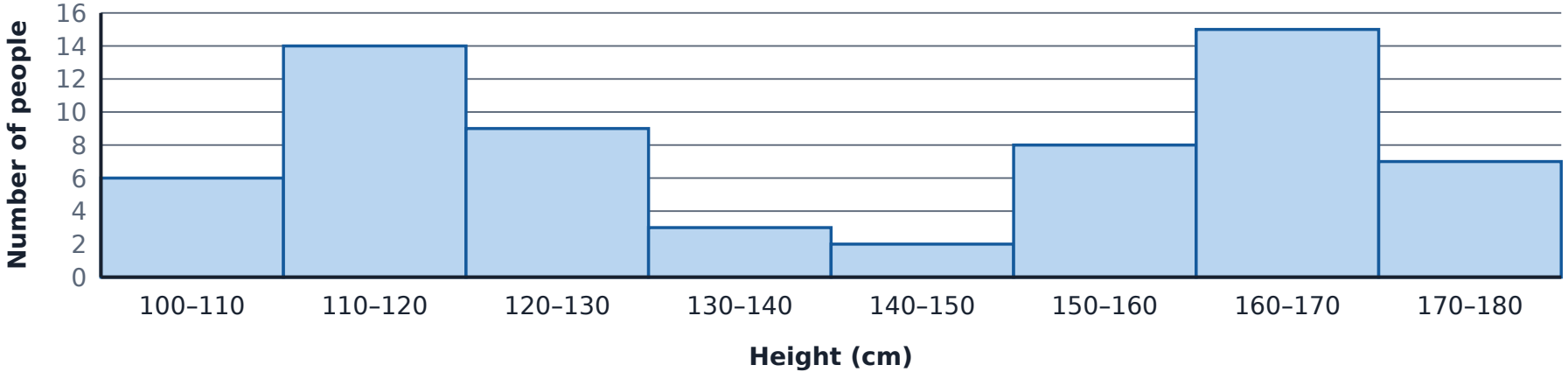
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q2/5

The histogram shows heights of all people at a school event (children and adults). Is a normal model appropriate? Justify.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q3/5

Waiting times at a clinic have mean 4 minutes, standard deviation 5 minutes, and minimum 0. Is a normal model appropriate? Justify using the numbers.

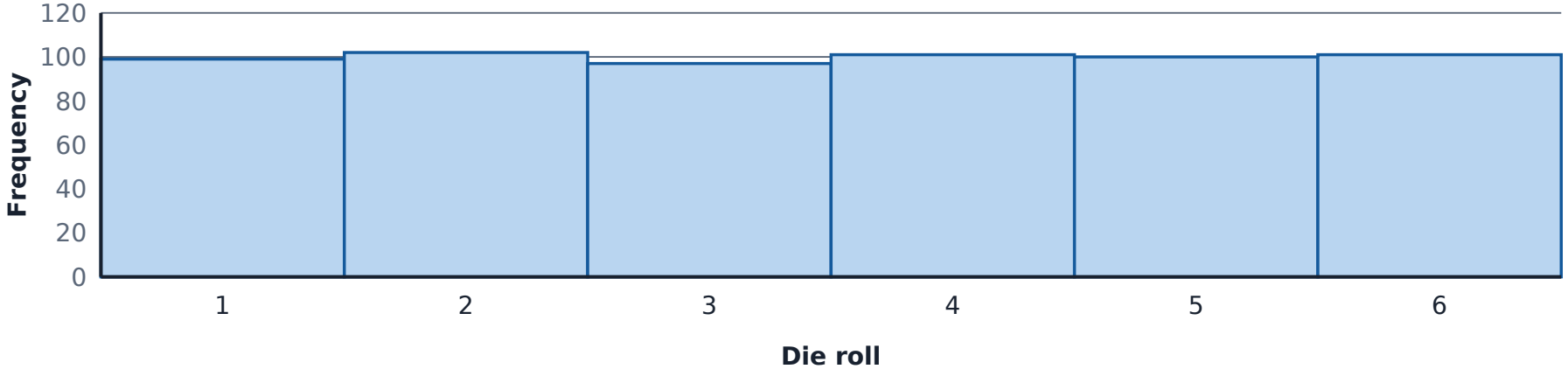
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q4/5

The histogram shows the results of 600 rolls of a fair die. Is a normal model appropriate? Justify.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 1

Describe skewed, symmetric, bimodal, and uniform shapes • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A histogram has a long tail to the right. Name its shape.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q1/5

A histogram has two separate peaks. Name its shape.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q2/5

True or false?

In a right-skewed distribution, the mean is usually less than the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q3/5

A histogram with bars all about the same height is

A. skewed left

B. uniform

C. bimodal

D. normal

Which is more resistant to a long tail: the mean or the median?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 2

Measures of center and spread for a data set • Grade 6 • Statistics & Probability • 6.SP.A.2

What does the standard deviation (or another spread measure) describe?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q1/5

Data are all at least 0 with mean 4. Can many values be far below 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q2/5

True or false?

A distribution can be described by its center, spread, and shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q3/5

Which is a measure of center?

A. median

B. range

C. standard deviation

D. IQR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q4/5

Give one value two standard deviations below a mean of 4 with SD 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 3

Groups combined in one data set • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Two groups with different centers are combined. What might the histogram show?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q1/5

Name a variable whose histogram for all students in a K-12 school might have several peaks.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q2/5

True or false?

A histogram of one group can look different from a histogram of two groups combined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q3/5

A bimodal histogram suggests

A. an error in counting

B. a small standard deviation

C. two groups in the data

D. a normal model

How could you get a mound-shaped distribution from the school-event heights?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q5/5

SET 62 • HSS-ID.A.4

FORWARD 1

Check a normal model with the 68-95-99.7 rule • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

In 1,000 measurements, 70% are within 1 SD of the mean and 96% within 2 SDs. Is a normal model reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q1/5

In another data set, 85% of values are within 1 SD and 91% within 2 SDs. Is a normal model reasonable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q2/5

True or false?

If a data set has an extreme outlier, the normal model may still fit the rest but the mean and SD are distorted.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q3/5

About what percent of normal data lies more than 3 SDs from the mean?

A. 5%

B. 3%

C. 0.3%

D. 32%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q4/5

A data set of 200 values has 25 values more than 2 SDs from the mean. Compare with a normal model.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSS-ID.A.4

FORWARD 2

Decide whether a probability model fits observed data • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A spinner is claimed to land on each of 4 colors equally often. In 400 spins the counts are 98, 105, 97, 100. Are the results consistent with the claim?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q1/5

A model says a coin is fair. In 50 flips there were 34 heads. In 200 simulated runs of 50 fair flips, 3 had 34 or more heads. Is the model consistent with the data?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q2/5

True or false?

A model is rejected only if the observed result would be unusual under the model.

A. True

B. False

A simulated result at least as extreme as the observed one occurs in 34 of 200 runs. The model is

A. proved true

B. rejected

C. consistent with the data

D. impossible

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q4/5

Explain the difference between judging a normal model for a data set and judging a probability model with a simulation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSS-IC.A.1 • STATISTICS, INFERENCE, AND DECISIONS

Distinguish population, sample, parameter, and statistic

MAIN	Algebra II main question	**HSS-IC.A.1**
BACKWARD 1	Samples and populations	**7.SP.A.1**
BACKWARD 2	Draw inferences from random samples	**7.SP.A.2**
BACKWARD 3	Compute sample proportions and means	**6.SP.B.5.c**
FORWARD 1	Describe sampling variability	**HSS-IC.A.1**
FORWARD 2	Quantify variability with a margin of error	**HSS-IC.B.4**

SET 63 • HSS-IC.A.1

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

A city surveys 400 randomly chosen residents, and 62% support a new park. Identify the population, the sample, the parameter, and the statistic.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q1/5

A factory measures 50 randomly selected bolts from a day's production; their mean length is 5.02 cm. Identify the parameter and the statistic.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q2/5

A news website posts a poll; 80% of the 2,000 visitors who answered want a later school start. Can this be generalized to all students? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q3/5

Explain why a random sample allows conclusions about the whole population.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q4/5

A school uses the first 30 students arriving in the morning to estimate the average hours of sleep. Is this a random sample? Explain a possible bias.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 1

Samples and populations • Grade 7 • Statistics & Probability • 7.SP.A.1

What is a population in statistics?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

What is a sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

True or false?

A random sample gives every member of the population a chance to be chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

Surveying every student in a school is

A. a statistic

B. an experiment

C. a sample

D. a census

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

Why are samples used instead of the whole population?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 2

Draw inferences from random samples • Grade 7 • Statistics & Probability • 7.SP.A.2

In a random sample of 50 students, 20 walk to school. Estimate how many of 600 students walk.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q1/5

Two random samples give 40% and 44%. Is that surprising?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q2/5

True or false?

Different random samples from the same population usually give slightly different results.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q3/5

From a random sample, 30% of 200 voters prefer a candidate. The best estimate for all voters is

A. 30%

B. 200

C. cannot estimate

D. 60%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q4/5

Why is a larger random sample usually better?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 3

Compute sample proportions and means • Grade 6 • Statistics & Probability • 6.SP.B.5.c

248 of 400 people say yes. What proportion is that?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q1/5

Find the mean of 5.0, 5.1, 4.9, 5.08.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q2/5

True or false?

A proportion can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q3/5

1,600 of 2,000 is

A. 16%

B. 1.25%

C. 80%

D. 20%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q4/5

Write 62% as a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q5/5

SET 63 • HSS-IC.A.1

FORWARD 1

Describe sampling variability • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Two random samples of 400 residents give 62% and 59%. Explain why they differ and what each estimates.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q1/5

Would samples of 40 or of 400 vary more from sample to sample? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q2/5

True or false?

A parameter changes from sample to sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q3/5

Which is a statistic?

A. the population size

B. the mean height of 30 sampled students

C. the mean height of all students

D. the population proportion

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q4/5

Explain why a large voluntary-response sample can still be worse than a small random sample.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q5/5

SET 63 • HSS-IC.A.1

FORWARD 2

Quantify variability with a margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

Simulated proportions from samples of 400 with $p = 0.62$ have SD 0.024. Give a margin of error.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q1/5

Use the margin of error to give an interval for the percent of residents who support the park.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q2/5

True or false?

A margin of error describes chance variation, not bias.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q3/5

If the sample size is multiplied by 4, the margin of error is about

A. unchanged

B. halved

C. doubled

D. divided by 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q4/5

Explain what the interval says about the population percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q5/5

SET 64 • HSS-IC.A.2 • STATISTICS, INFERENCE, AND DECISIONS

Decide whether a model is consistent with observed results using a simulation

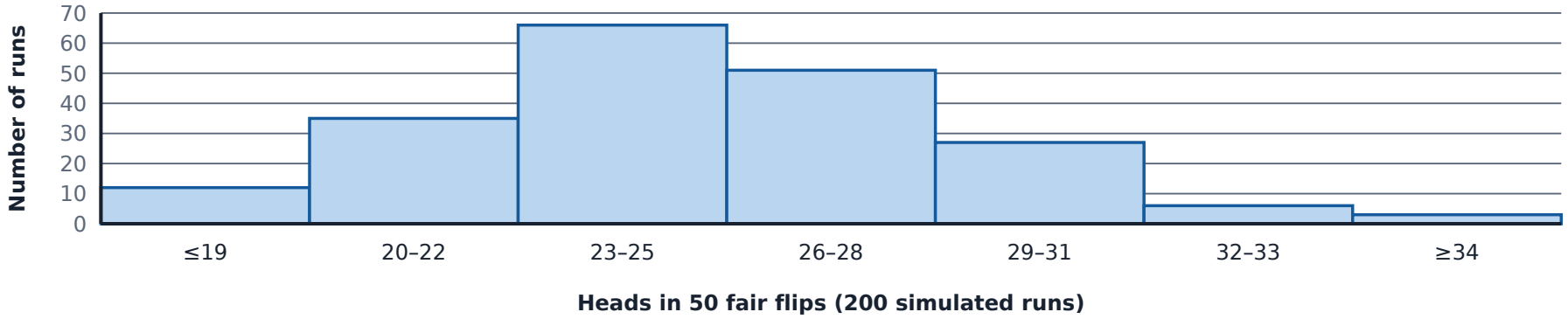
MAIN	Algebra II main question	**HSS-IC.A.2**
BACKWARD 1	Probability as long-run relative frequency	**7.SP.C.6**
BACKWARD 2	Use simulations of compound events	**7.SP.C.8.c**
BACKWARD 3	Relative frequency of a group of outcomes	**6.RP.A.3.c**
FORWARD 1	Design a simulation to test a model	**HSS-IC.A.2**
FORWARD 2	Compare two treatments with a re-randomization simulation	**HSS-IC.B.5**

SET 64 • HSS-IC.A.2

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A coin is claimed to be fair. It landed heads 34 times in 50 flips. The histogram shows 200 simulated runs of 50 fair flips. Is the claim consistent with the result? Justify with the simulation.



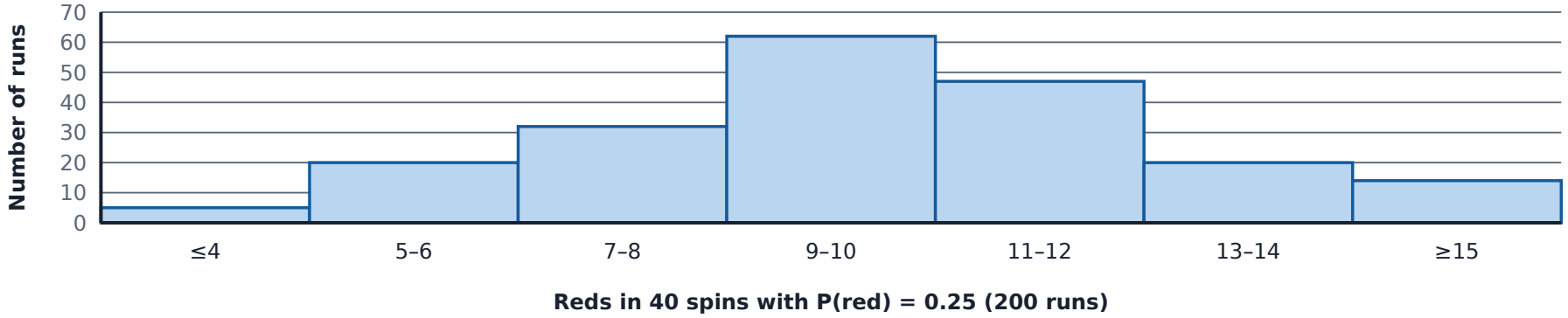
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q1/5

A spinner is claimed to land on red 25% of the time. It landed on red 13 times in 40 spins. The histogram shows 200 simulated runs. Is the claim consistent? Justify.



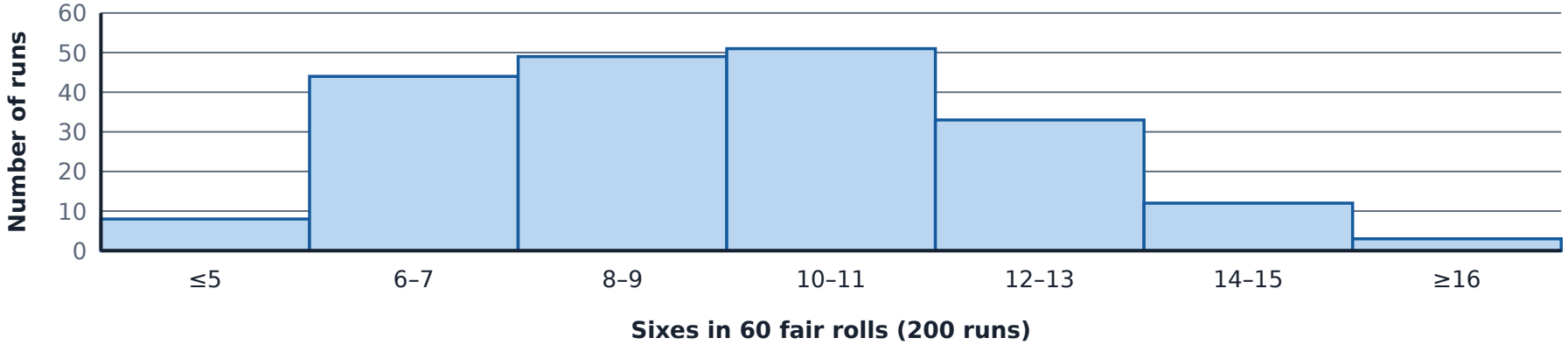
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q2/5

A die is claimed to be fair. It showed 16 sixes in 60 rolls. The histogram shows 200 simulated runs. Is the claim consistent? Justify.



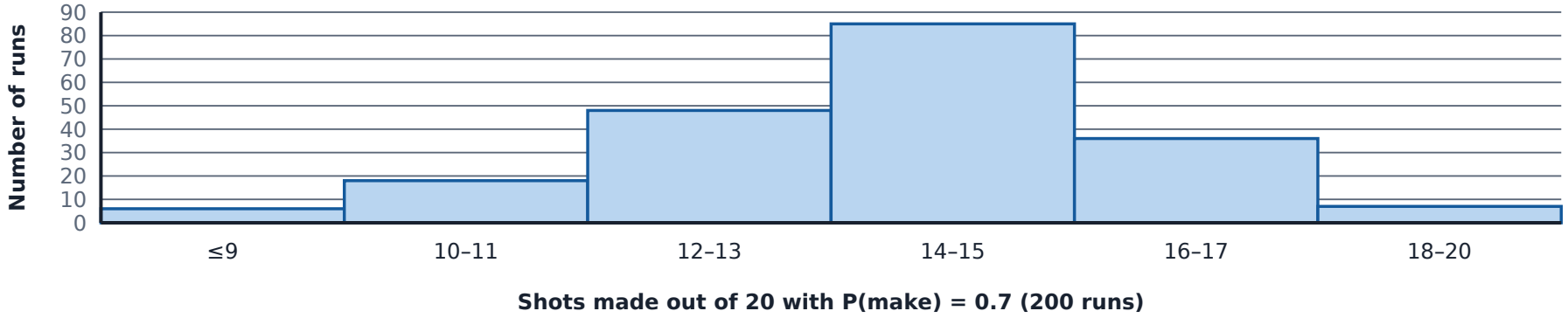
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q3/5

A player claims to make 70% of free throws. She made 11 of 20. The histogram shows 200 simulated runs of 20 shots. Is the claim consistent? Justify.



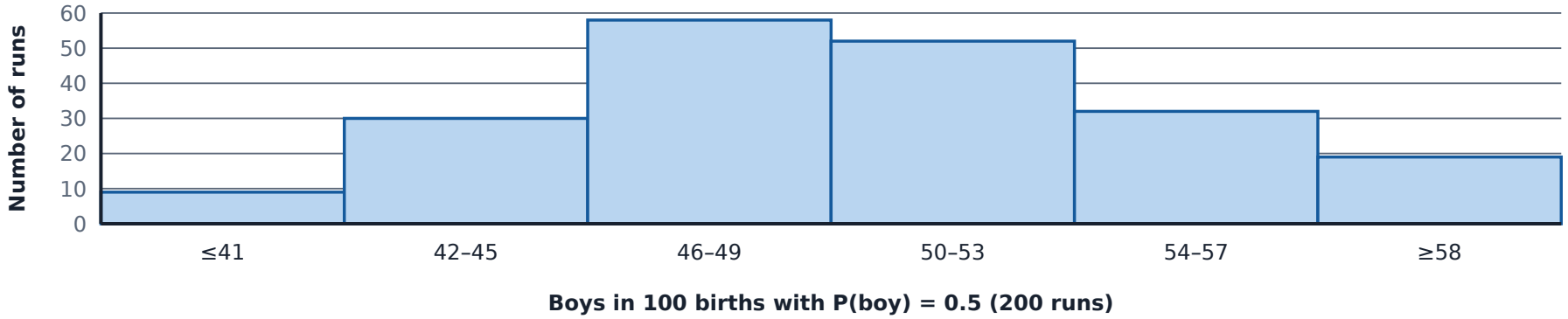
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q4/5

A model says boys and girls are equally likely. A hospital had 58 boys in 100 births. The histogram shows 200 simulated runs. Is the model consistent? Justify.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 1

Probability as long-run relative frequency • Grade 7 • Statistics & Probability • 7.SP.C.6

A fair coin is flipped 50 times. About how many heads do you expect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q1/5

A spinner has P(red) = 0.25. About how many reds in 40 spins?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q2/5

True or false?

Getting exactly the expected number every time is likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q3/5

A fair die is rolled 60 times. Expected sixes:

A. 10

B. 16

C. 6

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q4/5

Why is 26 heads in 50 flips not surprising?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 2

Use simulations of compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.c

Describe how to simulate 50 fair coin flips with random digits.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q1/5

How could you simulate one spin of a spinner with P(red) = 0.25 using two-digit random numbers 00-99?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q2/5

True or false?

Each run of a simulation should copy the real process under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q3/5

A simulation has 200 runs. Each run represents

A. the final answer

B. one repetition of the whole experiment

C. one histogram bar

D. one flip

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q4/5

Why are many runs used?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 3

Relative frequency of a group of outcomes • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

3 of 200 runs is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q1/5

6 + 18 runs out of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q2/5

True or false?

19 out of 200 is 19%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q3/5

34 of 200 is

A. 1.7%

B. 17%

C. 6.8%

D. 34%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q4/5

Is 2% less than 5%?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 3 • Q5/5

SET 64 • HSS-IC.A.2

FORWARD 1

Design a simulation to test a model • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

Describe a simulation to test whether a basketball player really makes 70% of free throws, given that she made 11 of 20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q1/5

A player who claims to make 70% of free throws made 11 of 20. Why should a simulation count results of 11 or fewer (as extreme or more extreme) rather than exactly 11?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q2/5

True or false?

A result that happens in 30% of simulated runs is strong evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q3/5

A common cutoff for "unusual" is

A. about 50% of runs

B. 100% of runs

C. about 5% of runs

D. about 25% of runs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q4/5

Doubling the number of runs from 200 to 400 changes what about the simulation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

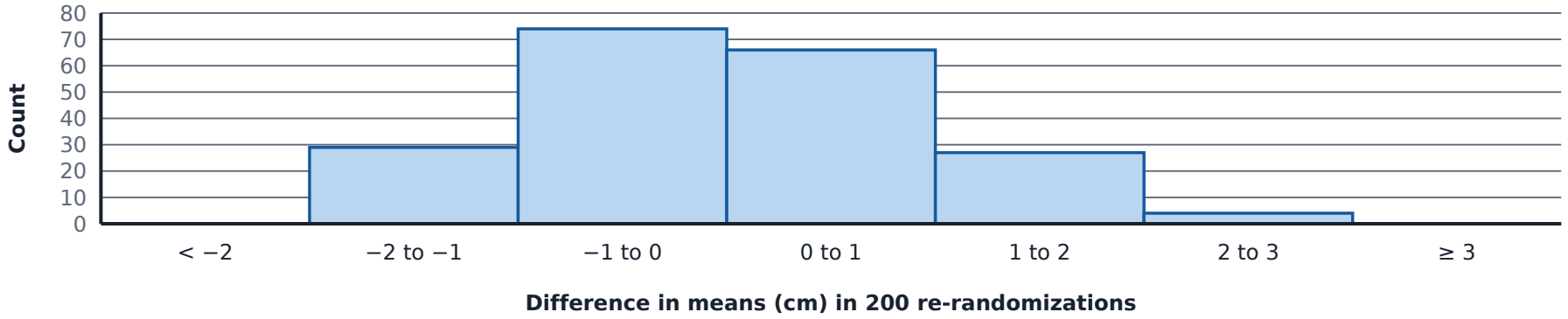
High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSS-IC.A.2

FORWARD 2

Compare two treatments with a re-randomization simulation • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In an experiment, fertilized plants grew 2.9 cm more on average. In 200 re-randomizations, none of the differences was 2.9 cm or more. What do you conclude?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q1/5

What model does the re-randomization assume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q2/5

True or false?

Re-randomization is a simulation of a model, like the coin-flip simulation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q3/5

If 60 of 200 re-randomized differences were at least the observed one, the result is

A. significant

B. proof of no effect

C. not statistically significant

D. impossible

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q4/5

Explain how a re-randomization test for an experiment extends the reasoning used to test a fair-coin claim with a simulation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 64 • FORWARD 2 • Q5/5

SET 65 • HSS-IC.B.3 • STATISTICS, INFERENCE, AND DECISIONS

Identify sample surveys, experiments, and observational studies

MAIN	Algebra II main question	**HSS-IC.B.3**
BACKWARD 1	Statistical questions and data collection	**6.SP.A.1**
BACKWARD 2	Association is not causation	**HSS-ID.C.9**
BACKWARD 3	Random samples	**7.SP.A.1**
FORWARD 1	Identify confounding and design flaws	**HSS-IC.B.3**
FORWARD 2	Analyze the results of an experiment	**HSS-IC.B.5**

SET 65 • HSS-IC.B.3

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers randomly assign 100 volunteers to take a vitamin or a placebo and compare colds. What type of study is this, and what is its purpose?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q1/5

A pollster asks a random sample of 1,000 voters which candidate they prefer. What type of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q2/5

Researchers compare the hospital records of smokers and nonsmokers for lung disease. What type of study is this? Can it show causation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q3/5

Which type of study best answers "Does playing music while studying improve test scores?" Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q4/5

Which type of study best answers "What percent of students at the school use the library weekly?" Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • MAIN • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 1

Statistical questions and data collection • Grade 6 • Statistics & Probability • 6.SP.A.1

Is "How many hours do students at our school sleep?" a statistical question? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q1/5

Give a data collection method for "What do voters think?"

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q2/5

True or false?

An experiment imposes a treatment on the subjects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q3/5

Watching and recording without imposing treatments is

A. an experiment

B. an observational study

C. a census

D. a placebo

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q4/5

What is a placebo?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 2

Association is not causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Ice cream sales and sunburns are correlated. Does ice cream cause sunburn? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q1/5

Name a lurking variable for "students with more books at home get higher scores".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q2/5

True or false?

A strong correlation proves that one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q3/5

A variable that affects both variables in a study is

A. the placebo

B. the sample

C. a confounding (lurking) variable

D. the response

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q4/5

What kind of study can show cause and effect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • HSS-IC.B.3

BACKWARD 3

Random samples • Grade 7 • Statistics & Probability • 7.SP.A.1